

ASML Holding N.V.

High-NA re-times, not reset; low-NA strength lifts forecasts

Reiterate Rating: BUY | PO: 1,710 EUR | Price: 1,305 EUR

High-NA adoption likely first in DRAM, last at TSMC

We discuss in this deep dive report the pros and cons of high-NA lithography. We conclude that DRAM vendors are likely to adopt high-NA first as they will not be impacted by the same stitching and design constraints that logic customers are dealing with. As such we expect Taiwan Semiconductor Manufacturing Company (TSMC) to adopt high-NA at the A10 node around 2032 while we think Samsung and SK Hynix could adopt it as early as 2028. We think Samsung Foundry and Intel will adopt high-NA before TSMC in order to differentiate vs TSMC with a possible focus on smaller die projects (mobile, PCs rather than AI logic).

Main challenges slowing high-NA adoption

While the merits of high-NA are clear on paper (better yields, lower costs), high-NA faces ongoing issues such as stitching (high-NA exposes only half the field of a low-NA tool), design/technology co-optimisation (high-NA brings new design rules), stochastics and inspection. These issues are particularly problematic for AI logic chips that are typically reticle-size (>800mm²). Further, changing the design rules creates risk for incumbents whose business model relies on strict rules around cell designs and transistor architecture. Equally these constraints are less problematic for DRAM vendors more focused on yields and whose chips are typically smaller. For foundry challengers, high-NA and chiplet designs could offer opportunities for market share gains, in our view.

Raising estimates and PO; reiterate Buy/top pick

We introduce in this note a bottom-up analysis of EUV adoption by customer based on capacity additions, layer count, yields and availability. This leads us to raise our C26-28E low-NA EUV tool deliveries from 62/81/75 to 65/85/87 and our EPS by 2.8-14.5% to €33.2/€46.5/€58.1 (3% to 20% above consensus). We have not changed our high-NA tool deliveries but have cut TSMC and increased other customers, consistent with our analysis. Our 2030 EPS sensitivity points to a range between €66 to €92 with a midpoint at €78 (26%). We reiterate our Buy rating/top pick and raise our PO to €1,710/US\$2,019 from €1,598/US\$1,886 on earnings upgrades with an unchanged methodology 30x CY27E EV/EBITDA, in line with its historical range of 18x to 34x and above its 26x median given accelerating growth (29% EBITDA CAGR '25 to '28E).

Estimates (Dec) (EUR)	2024A	2025A	2026E	2027E	2028E
EPS (Adjusted Diluted)	19.3	24.8	33.2	46.5	58.1
EPS Change (YoY)	-3.4%	28.4%	34.0%	40.1%	25.0%
Dividend / Share	6.48	7.50	8.70	10.6	13.0
ADR EPS (Adjusted Diluted - US\$)	20.9	28.0	38.9	54.4	68.1
ADR Dividend / Share (US\$)	7.01	8.46	10.2	12.4	15.2
Valuation (Dec)					
P/E	67.6x	52.7x	39.3x	28.1x	22.4x
Dividend Yield	0.50%	0.57%	0.67%	0.81%	1.00%
EV / EBITDA*	49.7x	40.1x	31.1x	22.8x	18.7x
Free Cash Flow Yield*	1.79%	2.17%	2.02%	3.95%	4.59%

* For full definitions of *IQmethod*SM measures, see page 37.

07 May 2026

Equity

Key Changes

(EUR)	Previous	Current
Price Obj.	1,598.00	1,710.00
2026E Rev (m)	39,149.7	39,935.2
2027E Rev (m)	46,988.7	49,712.8
2028E Rev (m)	52,001.0	57,774.6
2026E EPS	32.31	33.21
2027E EPS	43.20	46.51
2028E EPS	50.77	58.14

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Stock Data

Price (Common / ADR)	1,305 EUR / 1,545 USD
Price Objective	1,710 EUR / 2,019 USD
Date Established	7-May-2026 / 7-May-2026
Investment Opinion	B-1-7 / B-1-7
52-Week Range	587.80 EUR-1,313 EUR
Market Value (mn)	505,427 EUR
Shares Outstanding (mn)	387.3 / 387.3
Average Daily Value (mn)	1,214 USD
Free Float	97.8%
BofA Ticker / Exchange	ASMLF / ENA
BofA Ticker / Exchange	ASML / NAS
Bloomberg / Reuters	ASML NA / ASMLAS
ROE (2026E)	58.7%
Net Dbt to Eqty (Dec-2025A)	-54.1%

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Refer to important disclosures on page 38 to 41. Analyst Certification on page 36. Price Objective Basis/Risk on page 36.

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Timestamp: 07 May 2026 12:00AM EDT

iQprofileSM ASML Holding N.V.

Key Income Statement Data (Dec)

(EUR Millions)

	2024A	2025A	2026E	2027E	2028E
Sales	28,263	32,667	39,935	49,713	57,775
EBITDA Adjusted	10,021	12,408	16,010	21,801	26,611
Depreciation & Amortization	(919)	(1,026)	(1,054)	(1,100)	(1,155)
EBIT Adjusted	9,103	11,382	14,955	20,701	25,456
Net Interest & Other Income	19.8	105	136	125	150
Tax Expense / Benefit	(1,681)	(2,013)	(2,555)	(3,527)	(4,339)
Net Income (Adjusted)	7,599	9,639	12,746	17,566	21,565
Average Fully Diluted Shares Outstanding	394	389	384	378	371

Key Cash Flow Statement Data

Net Income (Reported)	7,546	9,606	12,717	17,543	21,563
Depreciation & Amortization	919	1,026	1,054	1,100	1,155
Change in Working Capital	2,053	1,027	(2,216)	2,164	1,046
Deferred Taxation Charge	0	0	0	0	0
Other CFO	648	1,000	(13.6)	40.0	40.0
Cash Flow from Operations	11,166	12,659	11,541	20,848	23,804
Capital Expenditure	(1,983)	(1,631)	(1,422)	(1,300)	(1,500)
(Acquisition) / Disposal of Investments	(626)	(2,147)	8.00	0	0
Other CFI	0	0	0	0	0
Cash Flow from Investing	(2,609)	(3,778)	(1,414)	(1,300)	(1,500)
Share Issue / (Repurchase)	(500)	(4,078)	(5,900)	(9,000)	(9,450)
Cost of Dividends Paid	(2,453)	(2,550)	(2,689)	(3,751)	(4,822)
Increase (decrease) debt	(954)	(968)	(3.40)	0	0
Other CFF	1,075	(1,074)	(655)	0	0
Cash Flow from Financing	(2,832)	(8,671)	(9,247)	(12,751)	(14,272)
Total Cash Flow (CFO + CFI + CFF)	5,725	210	880	6,797	8,032
FX and other changes to cash	6.50	370	(70.8)	0	0
Change in Cash	5,731	581	809	6,797	8,032
Change in Net Debt	(6,685)	(1,549)	(812)	(6,797)	(8,032)
Net Debt	(9,064)	(10,613)	(11,425)	(18,223)	(26,254)

Key Balance Sheet Data

Property, Plant & Equipment	6,847	7,894	7,975	8,175	8,520
Goodwill	4,589	4,589	4,589	4,589	4,589
Other Intangibles	621	540	532	532	532
Other Non-Current Assets	5,796	6,928	7,556	7,516	7,476
Trade Receivables	4,560	3,637	5,993	7,053	7,914
Cash & Equivalents	12,741	13,322	14,131	20,928	28,960
Other Current Assets	13,436	13,658	16,233	18,102	19,518
Total Assets	48,590	50,567	57,008	66,895	77,509
Long-Term Debt	3,677	2,709	2,706	2,706	2,706
Other Non-Current Liabilities	6,384	3,982	4,237	4,237	4,237
Short-Term Debt	0	0	0	0	0
Other Current Liabilities	20,051	24,264	26,247	31,342	34,665
Total Liabilities	30,113	30,954	33,190	38,285	41,608
Total Equity	18,477	19,612	23,817	28,610	35,901
Total Equity & Liabilities	48,590	50,567	57,008	66,895	77,509

Business Performance*

Return On Capital Employed	28.4%	34.5%	43.9%	52.1%	54.2%
Return On Equity	47.6%	50.6%	58.7%	67.0%	66.9%
Operating Margin	31.9%	34.6%	37.2%	41.5%	43.9%
Free Cash Flow (MM)	9,183	11,027	10,119	19,548	22,304

Quality of Earnings*

Cash Realization Ratio	1.47x	1.31x	0.91x	1.19x	1.10x
Asset Replacement Ratio	2.16x	1.59x	1.35x	1.18x	1.30x
Tax Rate	18.6%	17.7%	17.0%	17.0%	17.0%
Net Debt/Equity	-49.1%	-54.1%	-48.0%	-63.7%	-73.1%
Interest Cover	NA	NA	NA	NA	NA

* For full definitions of iQmethodSM measures, see page 37.

Company Sector

Semiconductor Capital Equipment

Company Description

ASML is the market leader in lithography tools, a critical part of the semiconductor manufacturing process enabling 'Moore's law'. The company benefits from technology transitions as well as new additions in leading edge logic and memory chip capacity. ASML currently has a market share of close to 90%, which is a monopoly in next-generation EUV lithography.

Investment Rationale

ASML has industrialised next gen EUV (Extreme Ultraviolet) lithography technology, which we believe will underpin many of the disruptive trends of this decade. Our Buy rating is predicated on 1) our long-term forecasts calling for double-digit revenue CAGR and upper-teen EPS CAGR over the next 5 years, and 2) potential upside to ASML's 2030 targets from AI and stronger industry demand.

Stock Data

Shares / ADR	1.00
Price to Book Value	21.0x

Quarterly Earnings Estimates

	2025	2026
Q1	6.03A	7.14A
Q2	5.90A	6.88E
Q3	5.49A	8.63E
Q4	7.37A	10.56E

Key Changes

(US\$)	Previous	Current
Price Obj.	1,886.00	2,019.00
2026E EPS	37.82	38.87
2027E EPS	50.57	54.44
2028E EPS	59.43	68.05

Contents

Investment takeaways	4
Portfolio manager summary	6
High-NA EUV: what really changes, and why that slows adoption	9
Bottom-up model: Low-NA EUV shipments by customer)	11
Bottom-up model: High-NA EUV shipments	13
What is High-NA EUV and how does it differ from Low-NA EUV?	15
Benefits of High-NA EUV versus Low-NA EUV	19
Challenges slowing High-NA adoption	23
Why TSMC is not committing (yet)	29
Why Intel, SK Hynix and Samsung are probably more willing	31
Glossary of technical terms	33

Investment takeaways

High-NA EUV is inevitable but not investable in the current cycle, in our view. ASML's upside through 2028 will be driven primarily by Low-NA EUV volume, mix and productivity, not a High-NA ramp. We expect DRAM adopts it first and TSMC last.

- **What's different:** Our bottom-up customer model shows Low-NA EUV shipments rising to 87 units in 2028 (vs ~75 previously), driven by Logic re-acceleration and DRAM EUV intensity, not High-NA pull-in. Overall, we model Logic EUV capacity additions increasing from 102k wspm last year to 211k wspm in 2028 while DRAM capacity additions (on EUV) increasing from 81k to 166k wspm in 2028.

Exhibit 1: Low-NA EUV demand remains broad-based, with TSMC dominant but memory catching up

We raise our Low-NA EUV tool delivery forecasts from 62/81/75 to 65/85/87

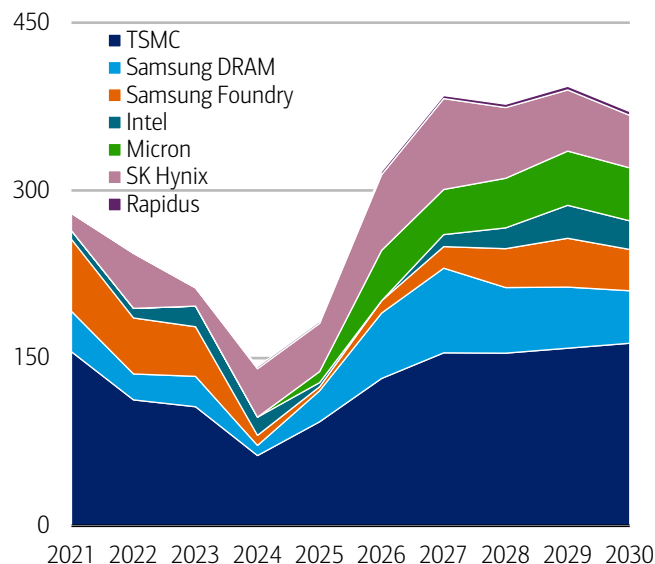
Low-NA	2021	2022	2023	2024	2025	2026	2027	2028
TSMC	23	21	26	14	19	28	34	35
Samsung DRAM	5	4	5	3	7	13	15	12
Samsung Foundry	10.0	8.0	10.0	3.0	2.0	4.0	6.0	10.0
Intel	2	3	9	9	2	0	4	6
Micron	0	0	0	0	3	5	10	11
SK Hynix	2	4	3	12	10	14	15	12
Rapidus	0	0	0	1	1	1	1	1
Total	42	40	53	42	44	65	85	87

Source: BofA Global Research estimates

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Exhibit 2: EUV capacity additions by customer

TSMC remains ASML's largest customer

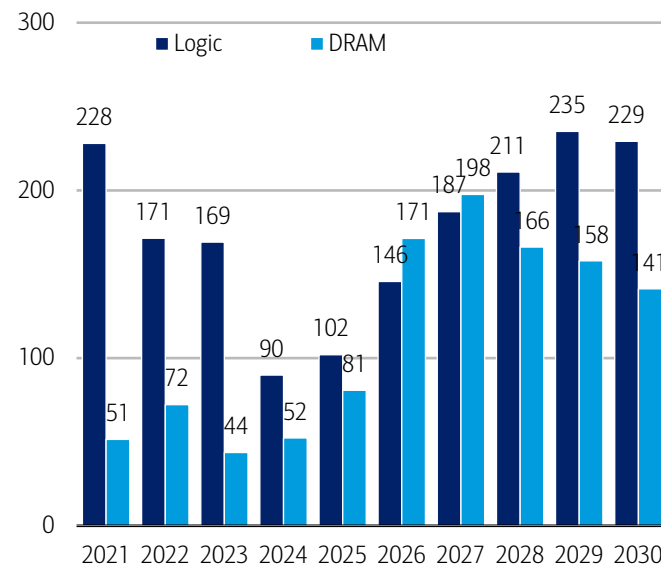


Source: BofA Global Research estimates

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Exhibit 3: EUV capacity additions by end-market

Capacity additions should peak in 2027 with clean rooms opening up



Source: BofA Global Research estimates

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- **High-NA reality check:** Stitching and DTCO reset make High-NA economically unattractive for platform foundries today. Adoption is layer-selective, memory-led, and back-end weighted.
- **Why ASML still wins:** ASML does not need High-NA to outperform. Low-NA combined with IBM already supports a multi-year growth runway; High-NA extends duration beyond the current cycle, in our view.
- **PO raised/ our Top Pick:** We reiterate our Buy / Top Pick; PO raised to €1,710 on earnings upgrades alone.

Exhibit 4: High-NA shipments by customer

We have reduced our high-NA tool shipments to TSMC and increased other customers

High-NA	2024	2025	2026	2027	2028	2029	2030
TSMC	0	1	1	2	2	2	3
Samsung DRAM	0	1	1	1	3	4	6
Samsung Foundry	0	0	0	1	1	2	2
Intel	2	1	1	1	3	5	4
Micron	0	0	0	1	1	1	1
SK Hynix	0	1	1	2	4	6	8
Rapidus	0	0	0	0	1	0	1
Total	2	4	4	8	15	20	25

Source: BofA Global Research estimates

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Exhibit 5: DRAM becomes the leading end-market for high-NA in 2028

Memory demand could reach 15 tools per year by 2030

High-NA	2024	2025	2026	2027	2028	2029	2030
Logic	2	2	2	4	7	9	10
DRAM	0	2	2	4	8	11	15
Total	2	4	4	8	15	20	25

Source: BofA Global Research estimates

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Portfolio manager summary

High-NA EUV: inevitable, but not investable - DRAM first, TSMC last

High-NA EUV delivers a clear optical upgrade versus Low-NA, but its economics and integration burden make adoption inherently uneven and slow, in our view. While higher numerical aperture (NA) improves resolution and reduces extreme multi-patterning on the most difficult layers, the shift to anamorphic optics introduces in-die stitching, new design rules and a material DTCO reset. These constraints are manageable for memory and selectively integrated logic designs, but structurally problematic for platform foundries serving large, reticle-scale SoCs. As a result, we expect High-NA adoption to be memory-led and layer-selective, with DRAM vendors adopting first and TSMC adopting last, around the A10 node (~2032). We believe Samsung Foundry and Intel are likely to engage earlier than TSMC to support differentiation and smaller-die logic programmes, but not at scale.

What actually drives ASML earnings: Low-NA EUV volumes, not High-NA timing

Our bottom-up, customer-level model reinforces our view that Low-NA EUV remains the primary earnings driver through the late 2020s. Capacity re-acceleration in advanced logic (AI-related compute, leading nodes) and rising EUV intensity in DRAM underpin higher Low-NA shipments, more than offsetting any deferral in High-NA adoption. We raise our Low-NA EUV unit forecasts to 65 / 85 / 87 tools in 2026–28E (from 62 / 81 / 75), reflecting higher wafer capacity additions, rising layer counts and improving productivity. High-NA volumes are unchanged near term and remain back-end weighted, with adoption constrained by stitching, DTCO friction, stochasticity and inspection rather than optical feasibility.

Estimates revision: higher revenues and EPS driven by Low-NA EUV volumes and mix

Reflecting higher Low-NA EUV volumes and a modest uplift in ArFi demand, we raise our revenue estimates by ~2% / ~6% / ~11% and EPS by +2.8% / +7.6% / +14.5% in 2026–28E to €33.2 / €46.5 / €58.1. Importantly, we make no change to High-NA tool volumes in the forecast period and no incremental margin assumptions. Gross margin expansion remains driven by EUV mix, Installed Base Management growth and operating leverage. Our framework implies ASML does not require an accelerated High-NA ramp to deliver strong earnings growth this cycle. Investment view: reiterate Buy / Top Pick; PO raised to €1,710 on unchanged methodology

Investment view: reiterate Buy / Top Pick; PO raised to €1,710 on earnings upgrades

We reiterate our Buy / Top Pick and raise our price objective to €1,710 (€2,019 ADR), based on an unchanged 30x CY27E EV/EBITDA, within ASML's historical range and justified by accelerating growth (EBITDA CAGR ~29% from 2025–28E), rising EUV layer count and ASML's reinforced monopoly position. Our updated 2030 EPS sensitivity (€66–€92; mid-point €78) highlights significant long-term optionality from High-NA without relying on early or aggressive adoption. In our view, High-NA extends the duration of ASML's growth profile rather than creating a near-term inflection; the bull case remains anchored in Low-NA EUV scale-up and services.

Exhibit 6: Summary of estimates change

Earnings upgrades driven by low-NA EUV volumes, not High-NA acceleration

€m	2026E			2027E			2028E		
	Old	New	% chg	Old	New	% chg	Old	New	% chg
DUV:									
Units	311.0	306.0	-1.6%	312.0	326.0	4.5%	340.0	355.0	4.4%
ASP (€m)	41.3	44.0	6.7%	41.0	43.4	5.7%	39.7	41.9	5.5%
Revenue	12,842.2	13,478.7	5.0%	12,803.7	14,136.3	10.4%	13,492.5	14,859.6	10.1%
Dry units	186.0	181.0	-2.7%	180.0	180.0	0.0%	205.0	205.0	0.0%
Dry ASPs (€m)	12.4	12.4	0.0%	12.5	12.5	0.0%	12.7	12.7	0.0%
Dry revenues (€m)	2,307.6	2,245.1	-2.7%	2,242.5	2,242.5	0.0%	2,607.5	2,607.5	0.0%
ArFi units	125.0	125.0	0.0%	132.0	146.0	10.6%	135.0	150.0	11.1%
ArFi ASP (€m)	84.3	89.9	6.6%	80.0	81.5	1.8%	80.6	81.7	1.3%
ArFi revenues (€m)	10,534.5	11,233.6	6.6%	10,561.2	11,893.8	12.6%	10,885.0	12,252.1	12.6%
EUV - Low-NA									
Units	62.0	65.0	4.8%	81.0	85.0	4.9%	75.0	87.0	16.0%
ASP (€m)	222.7	211.6	-5.0%	235.0	236.2	0.5%	250.0	253.0	1.2%
Revenue	13,804.4	13,753.4	-0.4%	19,035.0	20,076.5	5.5%	18,750.0	22,011.0	17.4%
EUV - High-NA									
Units	4.0	4.0	0.0%	8.0	8.0	0.0%	15.0	15.0	0.0%
ASP (€m)	450.0	450.0	0.0%	450.0	450.0	0.0%	450.0	500.0	11.1%
Revenue	1,800.0	1,800.0	0.0%	3,600.0	3,600.0	0.0%	6,750.0	7,500.0	11.1%
Total units	377.0	375.0	-0.5%	401.0	419.0	4.5%	430.0	457.0	6.3%
Blended	78.1	80.1	2.5%	91.1	92.9	1.9%	93.5	99.7	6.7%
Metrology and inspection rev.	1,015.6	1,015.6	0.0%	1,100.0	1,100.0	0.0%	1,200.0	1,200.0	0.0%
Total Product sales	29,462.2	30,047.7	2.0%	36,538.7	38,912.8	6.5%	40,192.5	45,570.6	13.4%
YoY growth (%)	20.4%	22.8%	2.4%	24.0%	29.5%	5.5%	10.0%	17.1%	7.1%
Installed base management	9,687.5	9,887.5	2.1%	10,450.0	10,800.0	3.3%	11,808.5	12,204.0	3.3%
YoY growth (%)	18.2%	20.7%	2.4%	7.9%	9.2%	1.4%	13.0%	13.0%	0.0%
Revenue	39,149.7	39,935.2	2.0%	46,988.7	49,712.8	5.8%	52,001.0	57,774.6	11.1%
YoY growth (%)	19.8%	22.2%	2.4%	20.0%	24.5%	4.5%	10.7%	16.2%	5.5%
Gross Profit	20,612.6	21,022.6	2.0%	25,600.2	27,081.7	5.8%	29,110.6	32,366.2	11.2%
Gross margin (%)	52.7%	52.6%	0.0%	54.5%	54.5%	0.0%	56.0%	56.0%	0.0%
Operating expenses	6,147.2	6,147.2	0.0%	6,460.4	6,460.4	0.0%	6,990.3	6,990.3	0.0%
EBIT	14,465.4	14,875.4	2.8%	19,139.8	20,621.3	7.7%	22,120.2	25,375.9	14.7%
EBIT margin (%)	36.9%	37.2%	0.3%	40.7%	41.5%	0.7%	42.5%	43.9%	1.4%
Adjusted:									
Basic EPS (€)	32.33	33.23	2.8%	43.03	46.32	7.6%	50.56	57.89	14.5%
Diluted EPS (€)	32.31	33.21	2.8%	43.20	46.51	7.6%	50.77	58.14	14.5%
DPS (€)	8.70	8.70	0.0%	10.60	10.60	0.0%	13.00	13.00	0.0%
Net cash (debt)	10,953.8	11,425.3	4.3%	15,853.9	18,222.7	14.9%	20,938.4	26,254.3	25.4%
Capex	-1,422.3	-1,422.3	0.0%	-1,300.0	-1,300.0	0.0%	-1,500.0	-1,500.0	0.0%
Bookings (total)	29,462.2	30,047.7	2.0%	38,365.6	40,858.5	6.5%	42,202.1	47,849.1	13.4%
Bookings (EUV)	16,348.8	16,293.3	-0.3%	24,898.5	26,044.2	4.6%	28,050.0	32,462.1	15.7%
Bookings (ex-EUV)	13,113.3	13,754.4	4.9%	13,467.1	14,814.3	10.0%	14,152.1	15,387.0	8.7%
FCF	9,647.5	10,119.1	4.9%	17,650.6	19,547.9	10.7%	19,356.8	22,303.8	15.2%

Source: BofA Global Research estimates

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Exhibit 7: 2030 scenario framework – lithography TAM and ASML revenue sensitivity

High-NA expands long-term optionality without being required to hit 2030 targets

€bn, unless otherwise stated	2030			Mid-point 2025-30E CAGR	Notes
Global Semis sales (US\$bn)	2,000	2,100	2,200	22%	
WFE (US\$bn)	248.0	265.7	286.0	18%	
WFE intensity	12.4%	12.7%	13.0%		Historical range 11.8% to 16.6%
Litho TAM (US\$bn)	60.8	66.4	72.9	18%	Assumes 25% litho intensity
Litho intensity	24.5%	25.0%	25.5%		
Litho TAM (€bn)	52.4	57.3	62.9		Assumes €/US\$ at 1.16
ASML litho revenues	48.5	53.0	58.2	17%	Assumes 92.5% share
o/w EUV	34.5	38.3	42.7	27%	Guide €22-32bn
EUV low-NA units	92	94	97		EUV capacity 90 units
EUV low-NA ASP	270	275	280		NXE:4200G productivity ≥ 280wph
EUV low-NA revenues	24.8	25.8	27.2	20%	
EUV high-NA units	20	25	30		Guide 25 units capacity in 2030
EUV high-NA ASP	485	500	515		EXE:5400D productivity ≥ 195wph
EUV high-NA revenues	9.7	12.5	15.5	57%	
o/w DUV	14.0	14.7	15.5	4%	Guide DUV + M&I €11-13bn
Dry units	164	150	143		
Dry ASP	14	14	14		Current ASP €14m
Dry revenues	2.3	2.1	2.0	3%	
ArFi units	130	140	150		
ArFi ASP	90	90	90		Current ASP €77m
ArFi revenues	11.7	12.6	13.5	4%	
Metrology	1.4	1.8	2.2	18%	
Total Systems sales	49.9	54.8	60.3	18%	
Installed Base Mgmt	15.6	17.9	20.4	17%	Guide €11-13bn
% of group sales	24%	25%	25%		10-year median 25.6%
Group revenues	65.4	72.7	80.8	17%	Guide €44-60bn
Gross profit	36.6	42.2	48.5		
Gross margin	56%	58%	60%		Guide 56-60%
OPEX	-8.7	-9.0	-9.4	9%	Guide €7.7-8.5bn
EBIT	27.9	33.2	39.1	24%	
EBIT margin	43%	46%	48%		
Net interest	0.2	0.2	0.2		
Profit before taxes	28.1	33.4	39.3		
Income tax	-4.8	-5.7	-6.7		Guide 17%
Equity income	0.3	0.3	0.3		
Net income	23.7	28.0	32.9	24%	
Share count (m)	357.8	357.8	357.8	-2%	
EPS	€ 66	€ 78	€ 92	26%	
2025-2030E CAGR	22%	26%	30%		

Source: BofA Global Research estimates

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High-NA EUV: what really changes, and why that slows adoption

High-NA EUV is an optical upgrade, but an integration downgrade

High-NA EUV increases numerical aperture from ~0.33 to ~0.55, delivering materially higher resolution and improved image contrast versus today's Low-NA EUV tools. In principle, this allows manufacturers to print tighter pitches with fewer patterning steps on the most challenging layers, relaxing k-factor pressure as EUV approaches its physical limits.

From a manufacturing economics perspective, however, High-NA is not a drop-in productivity upgrade. Achieving 0.55 NA requires a fundamentally different scanner architecture based on anamorphic optics, which reduces the exposure field size by roughly half. This shift changes the nature of lithography from a largely field-agnostic process to one that introduces in-die boundaries for larger designs.

As a result, High-NA simultaneously solves an optical constraint while introducing new system-level constraints. The key trade-off is not resolution versus cost alone, but resolution versus design and integration complexity. This distinction underpins the uneven adoption path we model across customers and end-markets.

Stitching and DTCO are the real bottlenecks, not scanner physics

The reduced exposure field inherent to High-NA means that large dies often require multiple exposures to complete a single layer, creating stitch boundaries inside the active circuitry rather than at the die edge. These stitched regions are highly sensitive to overlay, focus and dose variation, directly affecting yield and performance.

Crucially, this transforms High-NA from a lithography decision into a Design-Technology Co-Optimisation (DTCO) problem. Layouts must become stitch-aware; routing density near stitch regions must be constrained; standard-cell libraries and floorplans may need redesign. Unlike prior lithography transitions, this is not an incremental tightening of rules but a qualitative shift in design abstraction.

For vertically integrated manufacturers, this DTCO burden can be internalised and optimised for specific products. For a platform foundry serving hundreds of external customers, it represents a material cost in customer enablement, IP portability and learning-curve efficiency. In this context, we believe delaying High-NA adoption is not conservatism but an optimisation of return on design complexity.

This dynamic explains why High-NA adoption is inherently layer-selective and staged, rather than node-wide. The technology is viable, but its value only materialises on a narrow subset of the most resolution-limited layers, and only once the DTCO cost is justified by clear economic benefit.

Why memory adopts first and TSMC last

These integration constraints affect logic and memory very differently.

Modern DRAM dies are typically small (DDR5 die size is around 60-70 mm² while HBM3e/HBM4-class die size is around 95-120mm²) relative to the High-NA exposure field (c330-340mm²), allowing most memory layers to be printed without in-die stitching. In addition, resolution improvements map directly to cost-per-bit through tighter pitches and/or fewer patterning steps. As a result, High-NA offers a relatively clean economic proposition for advanced DRAM, particularly as EUV layer counts continue to rise in HBM-driven roadmaps.

By contrast, leading-edge logic dies – especially AI and HPC SoCs – are often reticle-scale (e.g. c850-860mm²). For these designs, stitching becomes unavoidable, and

the DTCO burden rises sharply. This disproportionately impacts platform foundries such as TSMC, whose competitive advantage rests on stable, scalable design rules and broad IP reuse across customers. Introducing High-NA prematurely risks fragmenting the platform for limited near-term gain.

Samsung Foundry and Intel sit between these two extremes, in our view. Their more vertically integrated models allow selective High-NA deployment for specific products (e.g. smaller-die logic, internal CPUs) as a means of differentiation, even if mass deployment remains gradual. This explains why we expect earlier experimentation and limited insertion at these customers, without a rapid volume ramp.

Overall, we see High-NA adoption is best understood not as a binary transition, but as a long-duration overlay on Low-NA EUV – complementary rather than substitutive. This framing is central to our shipment and earnings forecasts.

Bottom-up model: Low-NA EUV shipments by customer)

Framework: translate wafer capacity build to EUV tool demand

Our Low-NA model is built customer-by-customer and is intentionally operational: we start from annual wafer capacity additions (WSPM), translate that into EUV exposure demand using assumed EUV layer counts, then convert exposure demand into tool requirements using productivity (wph / kWPM equivalents), yield and an availability factor applied consistently across customers. In practice, this creates a coherent bridge between (i) the industry's capacity cycle and technology mix and (ii) ASML's shipment profile. The output of that exercise is the customer split shown in the shipment table (e.g., total Low-NA units rising to 65/85/87 in 2026–28E, with TSMC still the largest buyer but memory contributing a larger share versus earlier years). We use the same logic to reconcile the segment split (Logic vs DRAM) and the customer split, ensuring totals remain internally consistent. Note that many of our assumptions (especially yield and wafer capacity additions per chipmaker) are based on estimates.

Key technology inputs: EUV layer counts (per customer)

The model's most sensitive technology input is EUV layers per wafer, which we set by customer to reflect node position and product mix. For 2026–28E, we assume (illustrative highlights): TSMC 25/27/29 layers, Samsung Foundry 25/27/29, Intel 25/27/29, and Rapidus 25/27/29 (advanced logic-like intensity), while DRAM vendors run lower but rising intensity (Samsung DRAM 7/8/9; SK Hynix 7/8/9) and Micron ramps from a lower base (1/5/7) given late EUV adoption at the 1-delta node. This layer framework is the mechanism through which we reflect the note's core message: EUV intensity (layers) is a structural demand driver, not just wafer starts. Importantly, the same layer assumptions are used when aggregating to the Logic vs DRAM view, so that the segment totals (and the customer totals) reconcile mechanically rather than directionally.

Conversion to tools: productivity, yields, availability + what drives WSPM 2026–28E

We then convert EUV exposure demand into tools using two operating parameters: productivity and yield, with an availability factor applied consistently (we reference availability explicitly in the framework but do not change it idiosyncratically by customer in the tables). For Low-NA productivity we assume a steady improvement in average throughput from 230/240/250 wph (2026–28E), which maps to 138/144/150 kWPM equivalents in the model tables. Yields are customer-specific and improve unevenly: TSMC is held at 85%, while Samsung DRAM moves 65%→70%→65%, Samsung Foundry 50%→60%→67%, Intel 50%→50%→60%, Micron 65%→70%→75%, SK Hynix 70%→75%→70%, Rapidus 50%→50%→60% (2026–28E).

Capacity context (WSPM adds): the model embeds an up-cycle in capacity adds in 2026–28E with total additions driven by both segments: Logic WSPM adds 146/187/211, while DRAM adds 171/198/166. In Logic, the increase is primarily carried by TSMC (132/155/154) alongside improving contributions from Samsung Foundry (11/19/35) and a re-acceleration at Intel (0/11/19) by 2027–28E. In DRAM, the expansion is driven by Samsung DRAM (58/76/59), SK Hynix (68/81/63) and Micron (45/40/44). These capacity additions, combined with rising EUV layer intensity and improving productivity, reconcile to the Low-NA shipment ramp by customer and the step-up in total units.

Exhibit 8: Low-NA EUV demand drivers – tools, layers, productivity and yields

Low-NA EUV headroom comes from productivity and yield, not just fleet size

Low-NA units	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC	23	21	26	14	19	28	34	35	37	39
Samsung DRAM	5	4	5	3	7	13	15	12	9	7
Samsung Foundry	10	8	10	3	2	4	6	10	12	10
Intel	2	3	9	9	2	0	4	6	9	8
Micron	0	0	0	0	3	5	10	11	11	10
SK Hynix	2	4	3	12	10	14	15	12	9	7
Rapidus	0	0	0	1	1	1	1	1	1	1
Total	42	40	53	42	44	65	85	87	88	82
EUV layers	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC	10	14	20	22	23	25	27	29	31	33
Samsung DRAM	1	2	3	3	5	7	8	9	9	10
Samsung Foundry	7	10	15	20	22	25	27	29	31	33
Intel	7	10	15	20	22	25	27	29	31	33
Micron	0	0	0	0	1	1	5	7	8	9
SK Hynix	1	1	3	5	6	7	8	9	9	10
Rapidus	7	10	15	20	22	25	27	29	31	33
Avg productivity (wph)	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC	150	160	170	200	220	230	240	250	260	270
Samsung DRAM	150	160	170	200	220	230	240	250	260	270
Samsung Foundry	150	160	170	200	220	230	240	250	260	270
Intel	150	160	170	200	220	230	240	250	260	270
Micron	150	160	170	200	220	230	240	250	260	270
SK Hynix	150	160	170	200	220	230	240	250	260	270
Rapidus	150	160	170	200	220	230	240	250	260	270
Avg productivity (kwpm)	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC	90	96	102	120	132	138	144	150	156	163
Samsung DRAM	90	96	102	120	132	138	144	150	156	163
Samsung Foundry	90	96	102	120	132	138	144	150	156	163
Intel	90	96	102	120	132	138	144	150	156	163
Micron	90	96	102	120	132	138	144	150	156	163
SK Hynix	90	96	102	120	132	138	144	150	156	163
Rapidus	90	96	102	120	132	138	144	150	156	163
Yields	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC	75%	78%	80%	82%	85%	85%	85%	85%	85%	85%
Samsung DRAM	80%	80%	80%	30%	50%	65%	70%	65%	70%	75%
Samsung Foundry	50%	65%	65%	50%	30%	50%	60%	67%	72%	75%
Intel	30%	30%	30%	30%	30%	50%	50%	60%	65%	65%
Micron	85%	85%	85%	30%	50%	65%	70%	75%	75%	75%
SK Hynix	85%	85%	80%	60%	65%	70%	75%	70%	70%	75%
Rapidus	30%	30%	30%	30%	30%	50%	50%	60%	65%	70%
EUV %	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Samsung DRAM	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%
Micron	0%	0%	0%	0%	5%	10%	20%	25%	30%	35%
SK Hynix	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%
WSPM additions	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC	156	113	106	63	93	132	155	154	159	163
Samsung DRAM	36	23	27	9	28	58	76	59	55	47
Samsung Foundry	64	50	44	9	4	11	19	35	44	37
Intel	8	9	18	16	4	0	11	19	30	26
Micron	0	0	0	0	10	45	40	44	48	47
SK Hynix	15	49	16	43	43	68	81	63	55	47
Rapidus	0	0	0	2	2	3	3	3	3	3
WSPM additions	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Logic	228	171	169	90	102	146	187	211	235	229
DRAM	51	72	44	52	81	171	198	166	158	141
Blended yields	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Logic	63%	68%	64%	58%	73%	80%	78%	78%	79%	80%
DRAM	81%	83%	80%	51%	57%	67%	72%	69%	71%	75%

Source: BofA Global Research estimates

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Bottom-up model: High-NA EUV shipments

Framework: High-NA is layer-selective, so we model it as an overlay on Low-NA

For High-NA we use the same bottom-up architecture (WSPM → layers → exposures → tools), but the economic logic is different: High-NA is assumed to be deployed selectively on the hardest layers and therefore ramps gradually rather than replacing Low-NA. The resulting segment profile shows High-NA starting small and then accelerating total High-NA units of 2/4/4/8/15/20/25 in 2024–30E, with the mix shifting toward DRAM as adoption broadens (DRAM 0/2/2/4/8/11/15 vs Logic 2/2/2/4/7/9/10 over the same period). This structure is consistent with the note’s strategic conclusion that memory adopts High-NA earlier because it is less exposed to stitching and platform DTCO friction than large-die logic.

High-NA “layer insertion” assumptions by customer

To keep High-NA grounded in manufacturing reality, we explicitly model how many High-NA layers are inserted rather than assuming whole-node conversion. The customer layer table shows High-NA layers are initially low (0–3 layers) even by 2030: for example, by 2030 the table indicates Samsung DRAM ~2 layers, Samsung Foundry ~3, Intel ~3, SK Hynix ~2, Rapidus ~2, while TSMC remains minimal (~2 by 2030 in the table). This is the mechanism through which we reflect “inevitable but not urgent”: High-NA contributes incremental tool demand without implying a near-term discontinuity in Low-NA shipments. We also include a High-NA mix indicator for memory (rising from 5% to 15% for Samsung DRAM and SK Hynix by 2030 in the table, with Micron later), consistent with an adoption curve that is memory-led and staged.

Conversion to tools: productivity, yields, availability + High-NA WSPM context (2026–28E)

High-NA tool counts are then derived using High-NA-specific operating assumptions for productivity and yields, again with an availability factor applied consistently (availability is part of the framework referenced in the note). The High-NA productivity table shows ~175/180/185 wph for logic-oriented adopters and ~190/190/195 wph for memory-oriented adopters (as listed in the model tables). High-NA yields are modelled lower / earlier in the learning curve and improve into 2030, with customer-specific trajectories shown (e.g., TSMC 30%→35%→40%, Samsung DRAM 50%→55%→60%, Samsung Foundry 20%→25%→40%, Intel 25%→30%→60% in the table’s ramp window).

Capacity context (WSPM adds): by construction, High-NA has no meaningful capacity contribution in 2026–27E in our framework; the first meaningful High-NA-linked WSPM contribution appears in 2028E, where the model shows Logic WSPM adds of 45 and DRAM adds of 18 associated with the early High-NA ramp window (with a much larger expansion thereafter). This helps explain why High-NA unit growth is back-half weighted while Low-NA remains the dominant near-term shipment driver.

Exhibit 9: High-NA EUV demand drivers – an economically selective ramp

High-NA adoption remains constrained by economics and integration complexity

High-NA units	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC				0	1	1	2	2	2	3
Samsung DRAM				0	1	1	1	3	4	6
Samsung Foundry				0	0	0	1	1	2	2
Intel				2	1	1	1	3	5	4
Micron				0	0	0	1	1	1	1
SK Hynix				0	1	1	2	4	6	8
Rapidus				0	0	0	0	1	0	1
Total	0	0	0	2	4	4	8	15	20	25
EUV layers	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC				0	0	0	0	0	0	2
Samsung DRAM				0	0	0	0	1	1	2
Samsung Foundry				0	0	0	0	2	2	3
Intel				0	0	0	0	2	2	3
Micron				0	0	0	0	0	0	1
SK Hynix				0	0	0	0	1	1	2
Rapidus				0	0	0	0	1	1	2
Avg productivity (wph)	2021	2022	2023	2024	2025	2026	2027	2028	2028	2028
TSMC								175	180	185
Samsung DRAM								190	190	195
Samsung Foundry								175	180	185
Intel								175	180	185
Micron								190	190	195
SK Hynix								190	190	195
Rapidus								175	180	185
Avg productivity (kwpm)	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC	0	0	0	0	0	0	0	96	98	101
Samsung DRAM	0	0	0	0	0	0	0	104	104	107
Samsung Foundry	0	0	0	0	0	0	0	96	98	101
Intel	0	0	0	0	0	0	0	96	98	101
Micron	0	0	0	0	0	0	0	104	104	107
SK Hynix	0	0	0	0	0	0	0	104	104	107
Rapidus	0	0	0	0	0	0	0	96	98	101
Yields	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC								30%	35%	40%
Samsung DRAM								50%	55%	60%
Samsung Foundry								20%	25%	40%
Intel								25%	30%	60%
Micron								50%	55%	60%
SK Hynix								50%	55%	60%
Rapidus								30%	35%	40%
High-NA	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Samsung DRAM	0%	0%	0%	0%	0%	0%	0%	5%	10%	15%
Micron	0%	0%	0%	0%	0%	0%	0%	0%	0%	5%
SK Hynix	0%	0%	0%	0%	0%	0%	0%	5%	10%	15%
WSPM additions	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
TSMC										61
Samsung DRAM								8	23	29
Samsung Foundry								10	25	27
Intel								36	74	81
Micron										3
SK Hynix								10	34	38
Rapidus										20
WSPM additions	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Logic	0	0	0	0	0	0	0	45	98	189
DRAM	0	0	0	0	0	0	0	18	57	70
Blended yields	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Logic								24%	29%	47%
DRAM								50%	55%	60%

Source: BofA Global Research estimates

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What is High-NA EUV and how does it differ from Low-NA EUV?

In this section, we discuss in more technical details the merits of high-NA lithography and its main advantages relative to low-NA EUV lithography as well as challenges to adoption.

Numerical aperture and why it matters

Numerical aperture (NA) is the single most important optical parameter in optical lithography because it defines how much of the diffracted light from the mask can be captured by the projection optics and therefore transferred onto the wafer. In simplified terms, lithographic resolution follows the Rayleigh equation, where the minimum printable feature size scales with wavelength divided by NA. In EUV lithography, the industry has already fixed wavelength at 13.5 nm, leaving NA as the primary remaining optical variable. Low-NA EUV scanners, such as ASML's NXE:3400, 3600 and 3800 platforms, operate at $NA \approx 0.33$. This has proven sufficient for scaling from 7 nm down to early 2 nm-class nodes by combining optics with increasingly sophisticated computational lithography and design-technology co-optimisation (DTCO).

In optical lithography, resolution is commonly described by the Rayleigh resolution equation:

$$\text{Critical Dimension (CD)} = k_1 \times \frac{\lambda}{NA}$$

where:

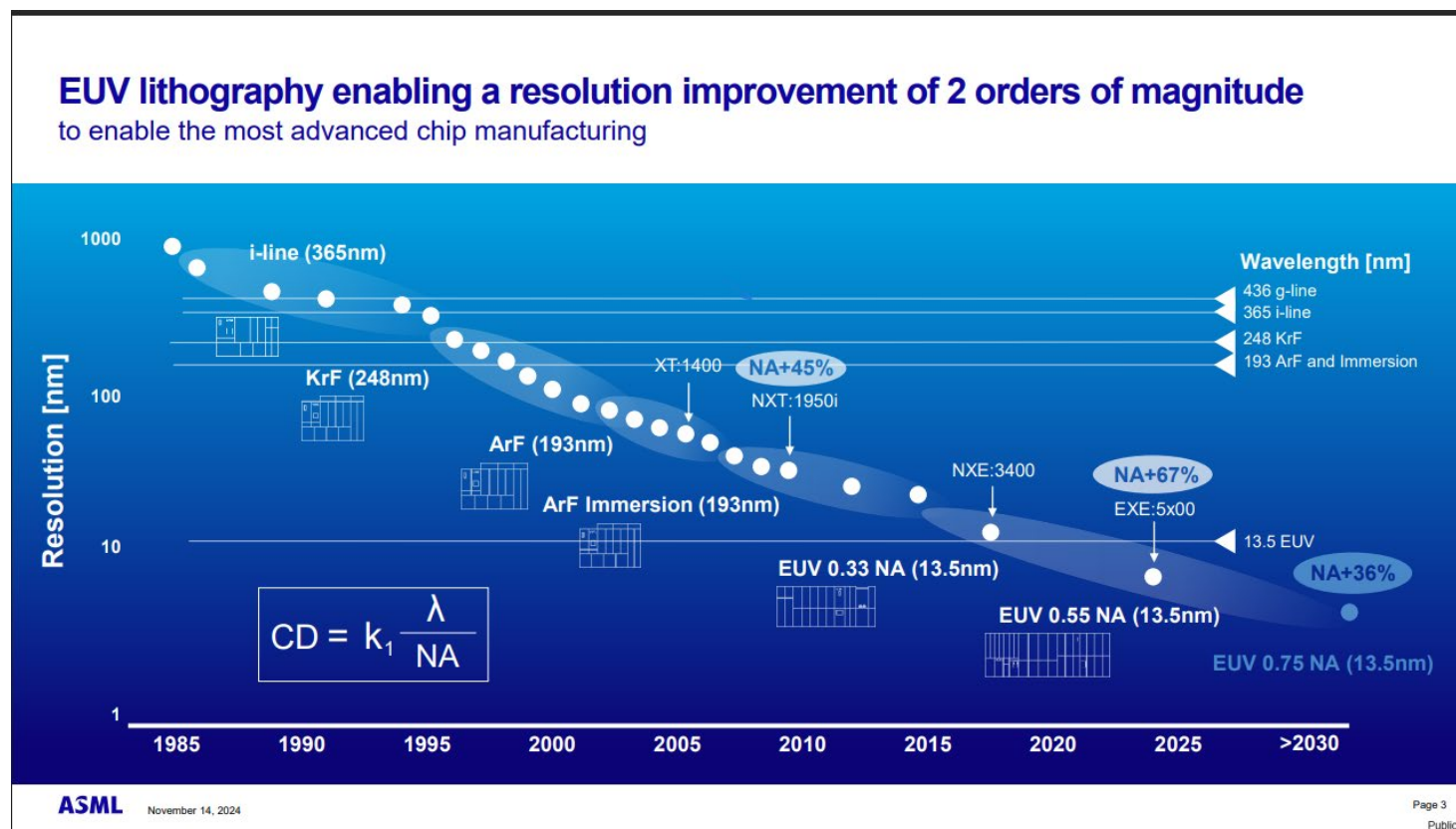
- λ (lambda) is the wavelength of light (fixed at 13.5nm for EUV),
- NA is the numerical aperture of the projection optics,
- k_1 (k factor) is a dimensionless process factor that captures everything else influencing resolution beyond wavelength and optics.

In practical terms, k_1 represents how aggressively the lithography process is being pushed. Early optical processes operated at $k_1 \approx 1.0$; modern EUV manufacturing typically runs close to $k_1 \approx 0.28$ – 0.30 , very near the physical limit of ~ 0.25 , below which usable image contrast collapses.

High-NA EUV represents a step change because it increases NA from 0.33 to approximately 0.55, which in optical terms is a very large jump. The physics implication is that more diffraction orders are captured by the lens system, enabling substantially finer pattern transfer with higher image contrast. This translates into roughly a 1.7× improvement in resolution and a corresponding reduction in k-factor dependence. From a manufacturing perspective, higher NA relaxes the need for extreme optical proximity correction (OPC) and multi-patterning to print small pitches, especially on the most critical layers. Importantly, this is not incremental progress along the existing EUV trajectory but a fundamental expansion of the optical process window, comparable in significance to the original transition from ArF immersion to EUV.

Exhibit 10: High-NA EUV (0.55 NA) enables much lower critical dimensions than low-NA EUV (0.33 NA)

ASML's roadmap shows how higher NA can reduce further feature size



Source: Company report

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For investors, we believe the relevance of NA lies in understanding that High-NA is not simply about “making features smaller”, but about giving the industry another degree of freedom once other scaling levers begin to saturate. As device architecture changes (FinFET to GAA to nanosheets) deliver diminishing returns, NA becomes an increasingly valuable knob. This explains why High-NA is widely viewed as inevitable even by customers currently delaying adoption. However, inevitability should not be confused with immediacy: the economic value of higher NA only materialises when resolution is the binding constraint, not merely a desirable improvement.

The anamorphic optics trade-off

To achieve NA ≈ 0.55 at EUV wavelengths, ASML had to adopt anamorphic projection optics, meaning the magnification differs in the scan (X) and slit (Y) directions. This is not a design preference but a physical necessity: at very high NA, symmetric optics would require lenses of impractical size and mass, with prohibitive distortion and photon loss. Anamorphic optics solve this by compressing the image differently along orthogonal axes, allowing the system to reach higher NA while maintaining optical performance and manufacturability. From a physics standpoint, this is an elegant solution to a hard constraint rather than a compromise chosen lightly.

The practical consequence of anamorphic optics is a reduced exposure field size on the wafer. In simple terms, each reticle exposure covers roughly half the field area of a low-NA EUV scanner (330-340mm² vs 850mm² for low-NA). While this does not affect small dies or test structures, it becomes structurally important for large logic dies, which can no longer be exposed in a single shot. Instead, multiple exposures must be “stitched” together within the same die. This requirement introduces a new class of lithographic boundary - inside the chip - rather than at die edges, which historically were less critical to electrical performance.

From an integration perspective, this is where High-NA departs most sharply from previous lithography transitions. All earlier transitions preserved a broadly similar exposure paradigm, allowing design abstractions to remain relatively stable. High-NA breaks that continuity. Stitching becomes a first-order design and yield consideration, not a secondary lithography issue. This trade-off - optical performance versus system-level complexity - is central to understanding why some customers may enthusiastically pursue High-NA while others, rationally, defer it.

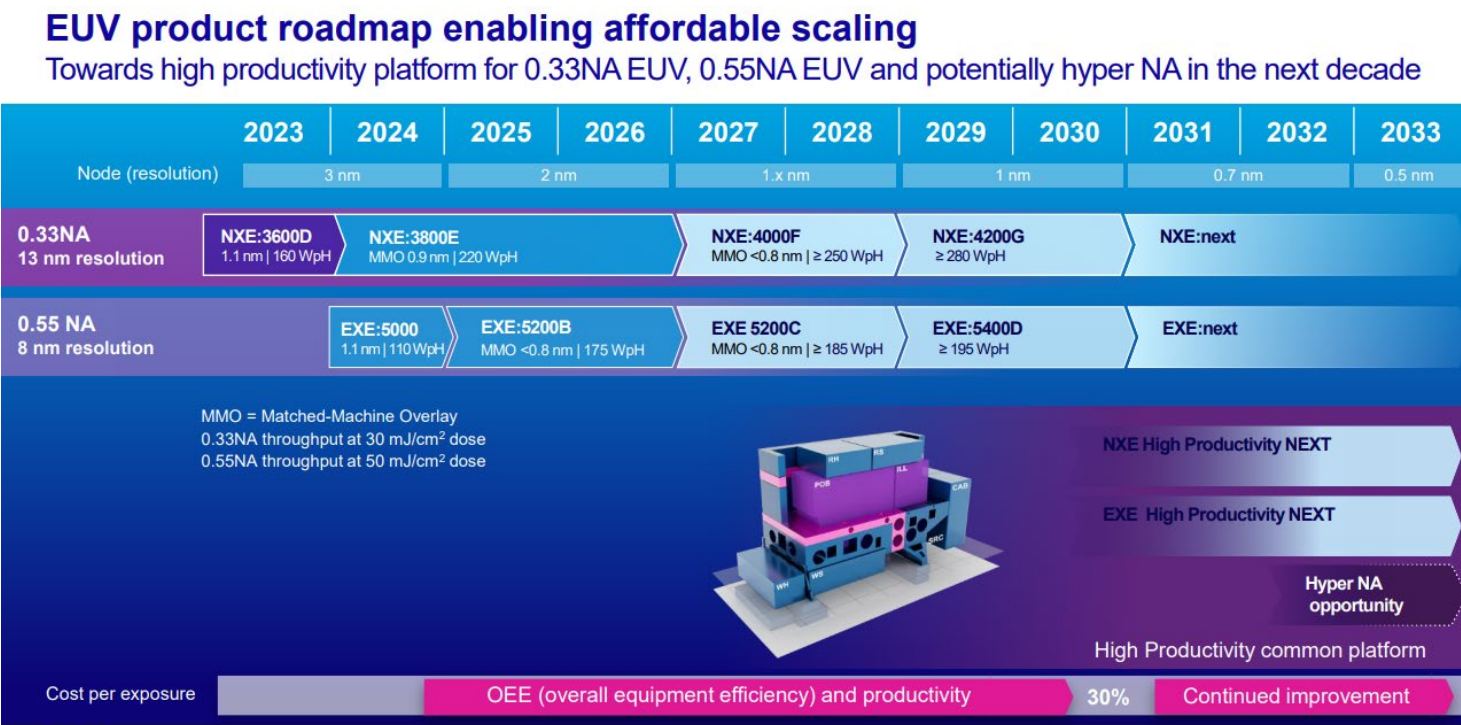
Platform implications for ASML

From ASML's perspective, High-NA EUV is best understood as a new lithography platform rather than an incremental extension of existing EUV tools. Achieving $NA \approx 0.55$ required deep redesign of virtually every subsystem: ultra-high-precision mirrors with tighter surface error tolerances, significantly heavier and stiffer wafer stages, more advanced vibration and thermal control, and upgraded control electronics capable of nanometre-scale stitching accuracy. These requirements are driven by physics and mechanical limits rather than optional feature sets, which explains the very high capital cost of High-NA systems.

This platform shift also implies that High-NA introduces new ecosystem dependencies. Masks must tolerate higher angular incidence, resists must balance sensitivity with line-edge roughness under higher photon densities, and metrology must resolve smaller features with tighter budgets. ASML therefore bears not just tool development risk but ecosystem coordination risk, similar in nature - though larger in scale - to the early EUV era. From a financial standpoint, this makes High-NA margin-accretive only once volumes scale and learning curves flatten.

For equity investors, the key point is that High-NA strengthens ASML's strategic monopoly while introducing temporal variability in revenue realisation. The platform reinforces ASML's long-term indispensability but does not require immediate broad adoption to be value-creating. We think high-NA should therefore be viewed less as a near-term growth driver and more as a long-duration option embedded in ASML's technology

Exhibit 11: ASML's EUV roadmap shows increasing productivity across both low-NA EUV and high-NA EUV
NXE and EXE are ASML's EUV platforms



Benefits of High-NA EUV versus Low-NA EUV

Resolution-driven simplification of patterning

The most direct manufacturing advantage of High-NA EUV is its ability to restore single-exposure patterning on layers where Low-NA EUV is approaching resolution limits. As critical pitches shrink, Low-NA increasingly relies on EUV double-patterning, cut masks, or highly aggressive OPC (Optical Proximity Correction), all of which increase process complexity. Each additional patterning step compounds overlay error, defect risk, and cycle time. High-NA's improved resolution and image contrast expand the process window, reducing the need for these workarounds on the most difficult layers.

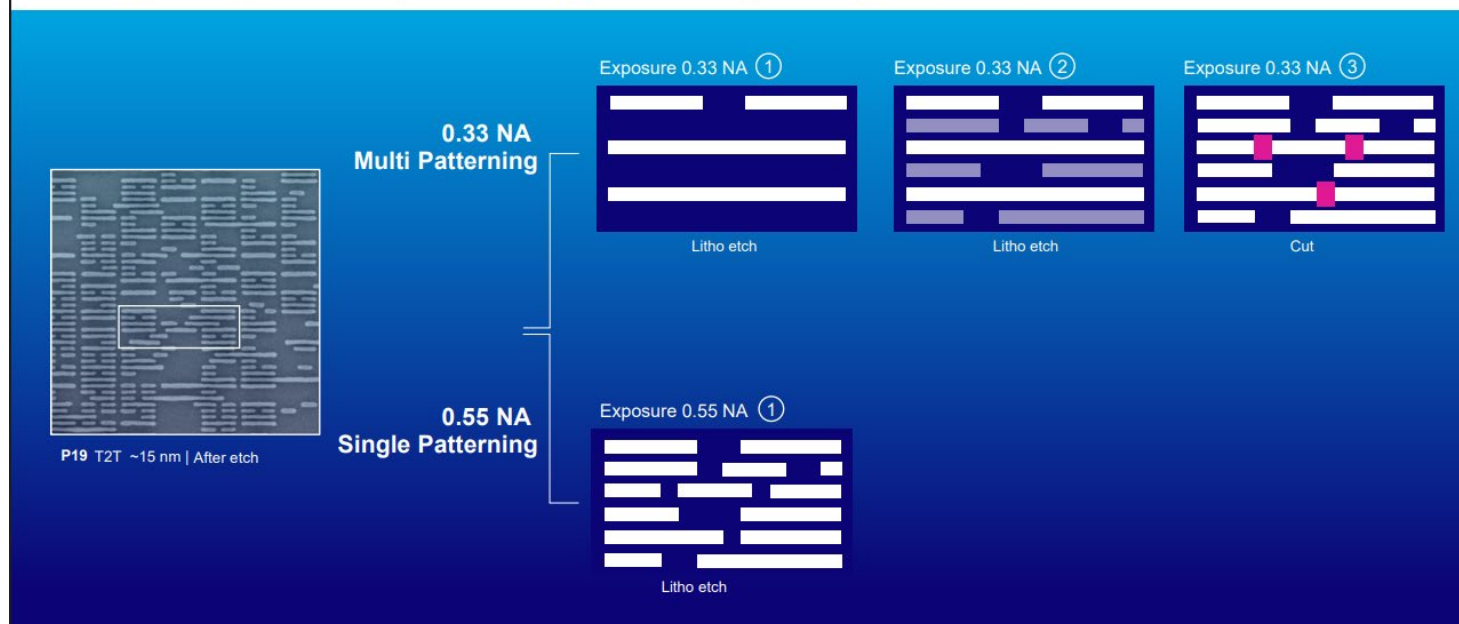
It is important to stress that this benefit is selective, not universal. Many layers in an advanced logic process are not resolution-limited and therefore gain little from High-NA. The economic justification for High-NA hinges on a relatively small subset of lithographically hardest layers - typically dense metal routing or critical junction layers - where process margin is otherwise exhausted. This layer-specific nature explains why customers can adopt High-NA experimentally without deploying it across an entire node.

For ASML customers, resolution-driven simplification matters less for headline node naming and more for yield stability and manufacturability. Where High-NA can eliminate marginal patterning steps, it improves not just yield but also operational resilience. This feeds into a broader calculus around cost of ownership and risk, rather than raw transistor density alone. According to ASML, high-NA EUV single exposure could enable c35% cost benefit vs multiple patterning low-NA EUV exposure.

Exhibit 12: High NA EUV single exposure significantly reduces process complexity vs low-NA EUV

ASML High NA EUV cost benefit

LOGIC [P19]: High NA EUV single exposure will enable ~35% cost benefit and significant process simplification compared to multi patterning 0.33 NA EUV



ASML | imec November 14, 2024

Cost benefit is calculated based on throughput and estimated non-litho cost.

Page 23
Public

Source: Company report

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Long-term control of EUV layer explosion

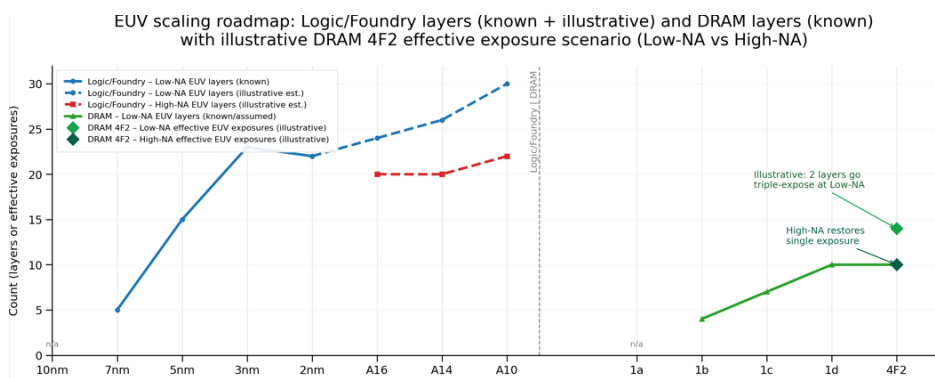
An emerging structural issue in advanced manufacturing is the rapid growth in EUV layer counts, particularly in the back-end-of-line and advanced memory flows. Each EUV layer adds exposure time, inspection load, resist cost and tool demand. Over time, the marginal benefit of adding EUV layers diminishes, while cost and complexity continue to rise. This phenomenon is especially acute in DRAM and HBM, where dense repeating structures push lithography harder each generation.

High-NA provides a mechanism to flatten this curve by enabling tighter pitches without multiplying patterning steps. Rather than adding more EUV layers to achieve scaling, manufacturers can use higher NA to extract more resolution from fewer exposures. From a physics standpoint, this is a more efficient use of photons and optics; from an economic standpoint, it reduces the growth rate of lithography intensity. Importantly, this benefit is cumulative over time rather than immediate, making it attractive for long-horizon process roadmaps.

For ASML, this dynamic supports a gradual High-NA ramp that aligns with increasing EUV intensity in memory and logic. High-NA does not replace Low-NA demand but complements it by preventing lithography cost from becoming prohibitive. Investors should therefore think of High-NA as a structural stabiliser for EUV economics rather than a step-function driver of wafer throughput.

Exhibit 13: High-NA adoption could reduce multiple patterning low-NA EUV in Logic and DRAM

High-NA promises a lower total cost of ownership for Logic and DRAM customers



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

2.3 Potential cost reduction, but only in specific cases

While High-NA is often framed as cost-saving, this is only true under specific conditions. High-NA tools are significantly more expensive than Low-NA EUV systems and initially deliver lower throughput due to tighter dose and focus requirements. Cost savings therefore depend on whether High-NA can replace multiple exposures or eliminate ancillary process steps. Where this substitution occurs, total wafer cost can indeed fall despite higher tool CAPEX.

However, if High-NA simply replaces a single Low-NA exposure, economics deteriorate. This is why High-NA adoption is inherently selective and often restricted to layers with extreme patterning requirements. The breakeven point depends on utilisation, yield impact and the maturity of the tool. Early adopters must be confident that High-NA will operate close to optimal conditions, which raises execution risk.

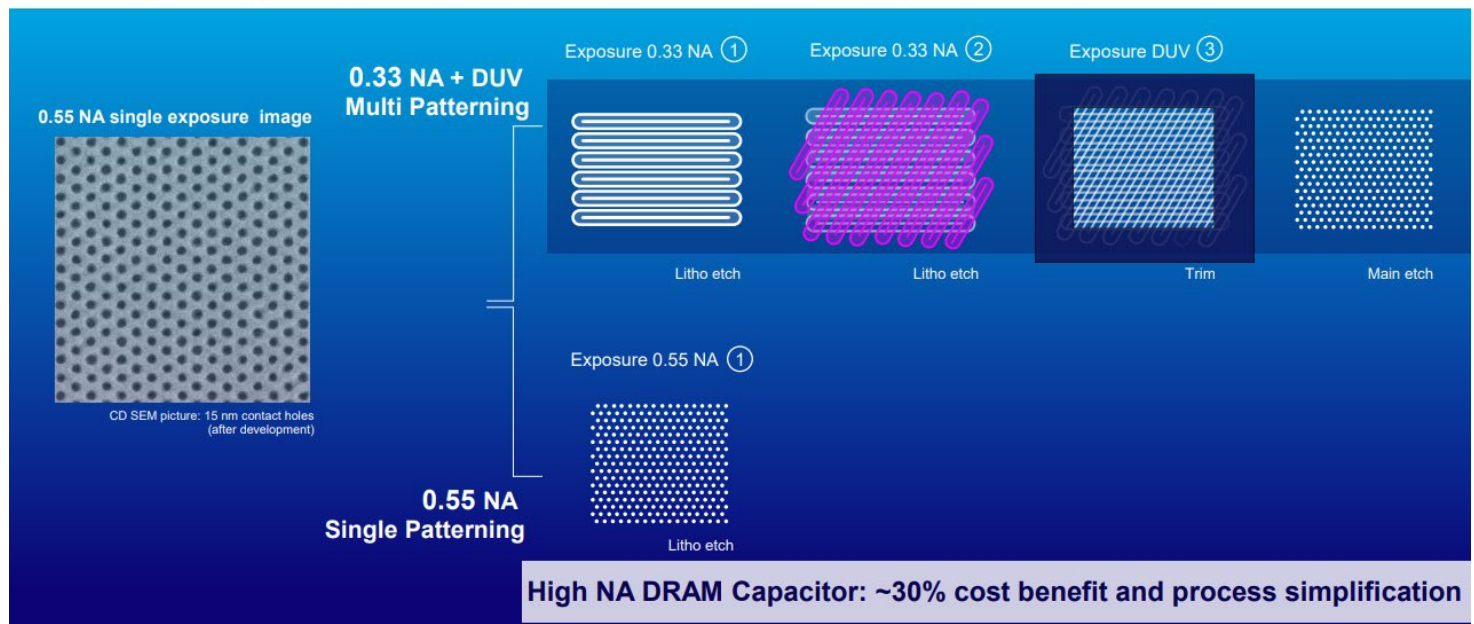
From an investor's perspective, this reinforces our view that High-NA revenue will build unevenly across customers and nodes. ASML's pricing power is strong, but customer economics ultimately dictate adoption speed. Cost reduction is possible, but it is not automatic.

Exhibit 14: High-NA EUV could reduce capital costs for DRAM companies

Moving from multiple patterning to single exposure eliminates a number of process steps

DRAM 15nm contact holes: High NA enables high productivity single exposure

Moving from triple exposure (2x 0.33NA EUV +1x DUV) to single exposure 0.55 NA EUV



ASML

November 14, 2024

Cost benefit is calculated based on throughput and estimated non-litho cost. | *depending on CD SEM settings and post processing

Page 26
Public

Source: Company report

BofA GLOBAL RESEARCH

Strategic optionality for ASML customers

Beyond near-term cost and yield metrics, High-NA provides strategic optionality as traditional scaling levers mature. Device architecture improvements have historically delivered large gains, but each successive innovation offers smaller incremental benefits. High-NA supplies a fundamentally different scaling vector that does not rely on redesigning transistors or interconnects.

This optionality is particularly valuable for leading manufacturers planning beyond the current node horizon. Even if High-NA is not deployed immediately, having it qualified and available de-risks future roadmaps. For ASML customers, we see High-NA functions as a hedge against slowing architectural innovation.

For ASML, we believe this reinforces long-term demand visibility. High-NA strengthens customer lock-in by ensuring that ASML remains essential even as other scaling paths mature.

Exhibit 15: ASML EUV tool portfolio (low-NA and high-NA)

Low-NA EUV tool ASPs have increased over time with higher productivity; high-NA (EXE: series) are materially more expensive

Platform	Status	Introduction	Resolution	MMO (nm)	Throughput (WpH)	Productivity gain	ASP €m	ASP gain
NXE:3600D	Released	2023	13 nm	≤ 1.1	≥ 160		165	
NXE:3800E	Released	2024	13 nm	≤ 0.9	≥ 220	38%	225	36%
NXE:3800F	Development	2027	13 nm	≤ 0.9	≥ 260	18%	260	16%
NXE:4200G	Development	2029	13 nm	≤ 0.8	≥ 300	15%	280	8%
NXE:4200H	Development	2031	13 nm	≤ 0.7	≥ 330	10%	300	7%
NXE:4600	R&D / HVM	2031	13 nm	n.a.	≥ 400	21%	355	18%

Platform	Status	Introduction	Resolution	MMO (nm)	Throughput (WpH)	Productivity gain	Throughput with stitching (WpH)	ASP €m	ASP gain
EXE:5000	Released	2023	8 nm	< 1.1	110		75	250	
EXE:5200B	Released	2025	8 nm	< 0.8	175	59%	135	400	60%
EXE:5200C	Development	2027	8 nm	< 0.8	190	9%	160	425	6%
EXE:5200D	Development	2029	8 nm	< 0.8	≥ 195	3%	≥ 175	430	1%
EXE:5400E	Development	2031	8 nm	< 0.7	≥ 210	8%	≥ 180	450	5%
EXE:5600	R&D	2032	8 nm	n.a.	≥ 250	19%	n.a.	500	11%

Source: BofA Global Research estimates, company report

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Challenges slowing High-NA adoption

This is the centre of gravity of the debate and the reason adoption is uneven.

Stitching: the single biggest structural issue

Stitching is arguably the most fundamental challenge of High-NA EUV. Because exposure fields are smaller, multiple exposures must be combined within a single die. The boundary where these fields meet is sensitive to overlay, focus and dose variation, creating localised pattern distortion risk.

Unlike traditional overlay errors at die edges, stitching errors occur within active circuitry, directly impacting yield and performance. Academic and EDA studies show that mitigating stitching requires special design rules, restricted routing near stitch boundaries, and stitch-aware cell libraries.

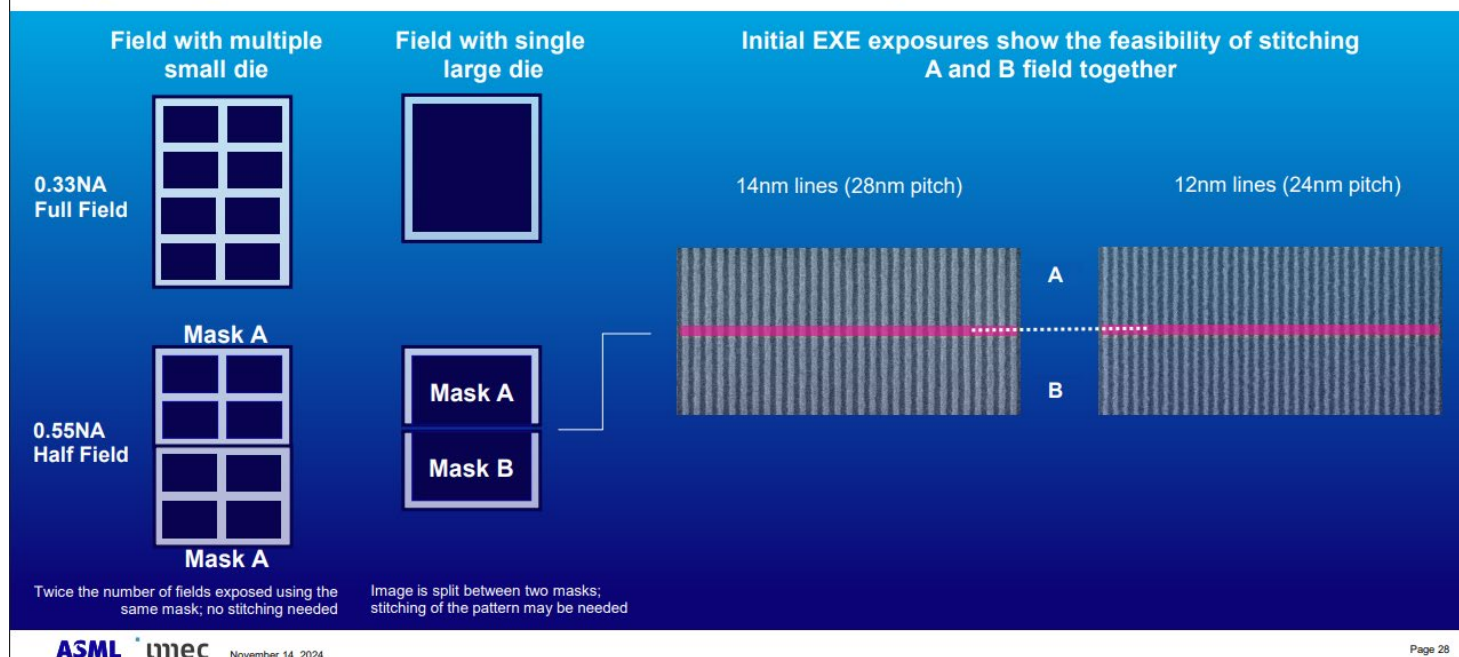
This transforms High-NA adoption from a lithography decision into a system-level integration challenge. Stitching is not a marginal inconvenience; it is a structural shift in how chips must be designed.

Exhibit 17: ASML is working on solving stitching issues for large die

High-NA could benefit from the industry's broad adoption of chiplets

High NA half field exposure requires stitching for large die

Engineering solutions available to enable at resolution stitching



Source: Company report

Page 28
Public

BofA GLOBAL RESEARCH

DTCO burden shifts upstream

Design-Technology Co-Optimisation (DTCO) refers to the tight, iterative co-development of device architecture, process technology, design rules, and physical layout to achieve manufacturable scaling beyond what lithography or devices alone would enable. As optical scaling has slowed, DTCO has become the dominant mechanism through which leading foundries continue delivering density, power and performance improvements. In practice, DTCO encompasses decisions such as transistor architecture selection (e.g. FinFET vs GAA), standard-cell height, routing pitch, track assignments, local interconnect schemes, and pattern regularity - all explicitly tuned to the strengths and limitations of

the manufacturing process. At advanced nodes, DTCO is no longer a downstream optimisation; it is the primary lever that makes the node economically and technically viable in the first place.

High-NA Turns DTCO from Field-Agnostic to Stitch-Aware

Under Low-NA EUV, DTCO has been progressively intensified but remains largely field-agnostic. That is, while layouts are increasingly restricted and regularised, designers can still assume a uniform exposure field without internal boundaries. High-NA EUV fundamentally breaks this assumption. Because High-NA uses anamorphic optics with a half-field exposure, layouts must now be stitch-aware: floorplans, placement, routing and even standard-cell libraries must explicitly avoid, tolerate, or manage stitch boundaries within the die. This shifts DTCO from optimising within a uniform process window to actively managing process discontinuities inside the chip. The DTCO burden therefore increases qualitatively, not incrementally. What was previously an optimisation problem becomes a constraint-management problem, with implications for yield, performance variability, and design effort.

High-NA DTCO Undermines IP Reuse at Platform Foundries

This distinction is critical to understanding TSMC's position. TSMC's competitive advantage is not simply superior transistor performance, but its ability to offer robust, scalable, and customer-agnostic platforms that support a very wide range of designs with predictable yields and fast learning curves. High-NA-specific DTCO - involving stitch-aware design rules, layer-selective constraints, and potential performance penalties near stitch regions - risks fragmenting that platform. It would require customers to adapt IP, redesign blocks, and accept node-specific layout caveats that undermine reuse and portability. For a vertically integrated manufacturer, such costs can be internalised. For a pure-play foundry serving hundreds of external customers, they represent real friction that must be justified by clear and immediate economic benefit.

Extending Low-NA DTCO Yields Higher Near-Term Returns

TSMC's strategy to date indicates that it prefers to extract the remaining value from Low-NA EUV by pushing DTCO further, rather than resetting the DTCO stack around High-NA prematurely. Advanced standard-cell architectures, aggressive design regularisation, and close alignment between routing and lithographic capabilities have so far allowed TSMC to meet A16 and A14 targets without High-NA. In this context, postponing High-NA is not technological conservatism but a rational optimisation of DTCO return on complexity. TSMC is effectively signalling that as long as DTCO on Low-NA continues to deliver platform-wide benefits, it is economically superior to adopting a technology that increases DTCO cost, complexity, and customer enablement risk for only marginal near-term gain.

Tool cost and throughput economics

High-NA EUV tools present a steep economic hurdle. Unit costs are roughly double those of Low-NA EUV (see table below), and early throughput is constrained by tighter process windows. Higher photon dose requirements further limit wafers-per-hour. Together, these factors increase depreciation per wafer at the outset.

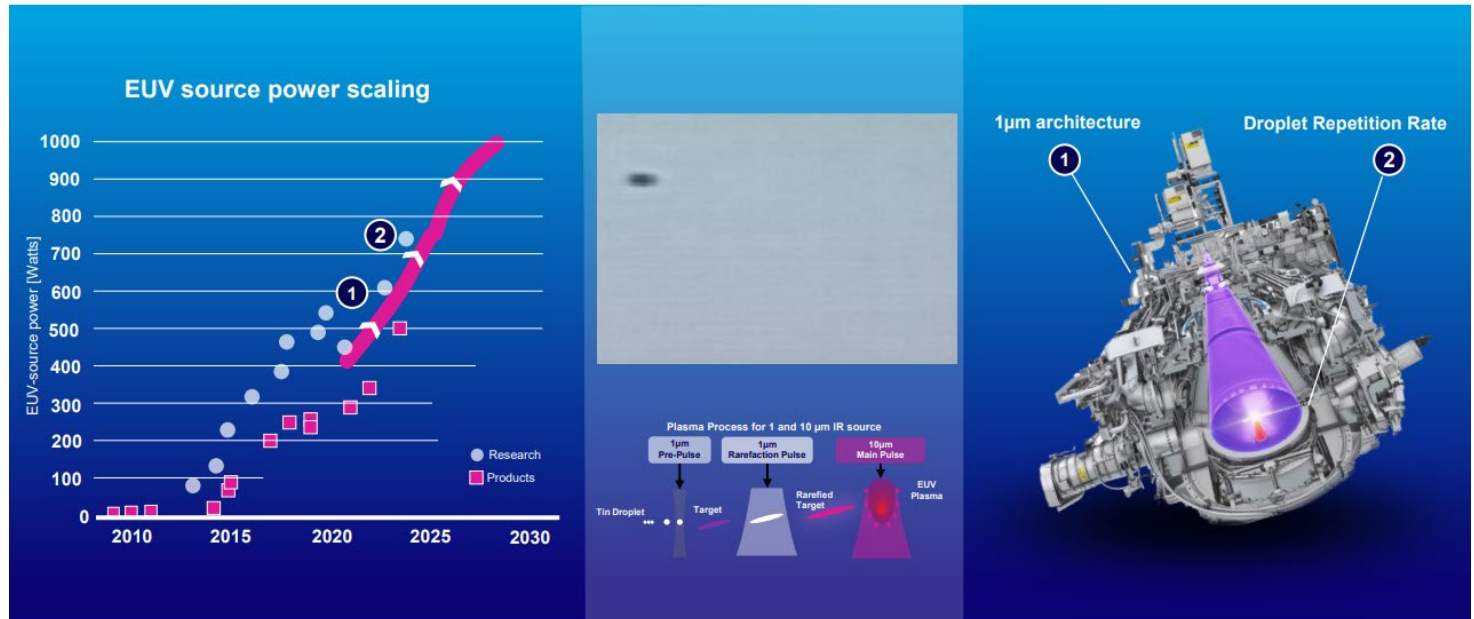
Customers with highly optimised Low-NA EUV operations therefore face a material opportunity cost in adopting High-NA too early. Even if High-NA is technically superior, it must outperform an already amortised, productive fleet to justify capital reallocation. This dynamic explains why finance organisations play a central role in adoption decisions.

Over time, throughput improvements are likely, but early adopters bear that risk. From an investor standpoint, this supports a measured adoption curve rather than an abrupt transition.

Exhibit 16: ASML has demonstrated substantial improvement in the source power with a roadmap to achieve >1000W performance by the end of the decade

Higher source power will help increase productivity (wafers exposed per hour) driving down costs for ASML customers

EUV source power continues to scale with opportunity to achieve >1000W 740W EUV power demonstrated - measures identified to reach >1000W in the future



ASML November 14, 2024

Page 31
Public

Source: Company report

BofA GLOBAL RESEARCH

EUV source power on track to reach >1,000W

A further constraint on High-NA adoption, often under-appreciated relative to stitching and DTCO, is EUV source power and the pace at which it can be scaled sustainably above 1,000W. Today, EUV productivity across both Low-NA and High-NA platforms remains fundamentally source-limited, not stage-limited. ASML has demonstrated EUV source power of approximately 740W, representing a >30× improvement versus early EUV systems and a clear validation of long-term scalability. Importantly, ASML has also articulated a credible roadmap to take EUV source power beyond 1,000W by the end of the decade, via improvements in droplet generation, CO₂ laser efficiency, debris mitigation and collector lifetime. However, this roadmap is non-linear: incremental gains in power become progressively harder to industrialise as thermal load, optics lifetime and availability trade-offs increase.

This matters disproportionately for High-NA economics. High-NA improves resolution, but it does so at the cost of tighter dose, focus and stochastic constraints, which tend to push required exposure dose higher than for comparable Low-NA layers. In other words, High-NA is structurally more sensitive to source power. At today's demonstrated EUV power levels, High-NA throughput remains meaningfully below mature Low-NA EUV, reinforcing the need for layer-selective rather than broad deployment.

Higher source power has got different benefits for low-NA vs high-NA

Reaching >1,000W EUV source power would be a genuine inflection, but its implications differ by platform:

- For **Low-NA EUV**, higher source power translates relatively cleanly into higher wafers-per-hour. The Low-NA optical platform is already mature, field sizes are full,

and many customers are running layers where process windows are well characterised. As a result, incremental source power largely converts into productivity and cost-per-wafer improvements, extending the economic life of the installed base and supporting the steady productivity gains assumed in our Low-NA model.

- For **High-NA EUV**, >1,000W is closer to a prerequisite than an optimisation. Higher source power is needed not only to lift throughput, but also to offset tighter stochastics, enable acceptable dose latitude, and stabilise yields on the most aggressive layers. Even at >1,000W, High-NA remains unlikely to match Low-NA productivity on a per-tool basis in the near term, but the gap narrows to a point where High-NA can begin to make economic sense on layers currently forced into multi-patterning under Low-NA EUV.

Higher power does not solve stitching, DTCO or inspection issues

Crucially, higher source power does not solve stitching, DTCO or inspection constraints. It improves the denominator of the cost-of-ownership equation, but not the numerator of design complexity. This is why we view source power scaling as a necessary enabler of broader High-NA adoption, but not a sufficient trigger for platform-wide insertion. Customers still require confidence that higher throughput does not come at the expense of optics lifetime, availability or system-level stability.

From an investor perspective, this reinforces two points in our thesis. First, Low-NA EUV remains the primary beneficiary of near-term source power gains, supporting continued productivity-driven upside without requiring High-NA volume acceleration. Second, High-NA's adoption curve remains gated by multiple system-level constraints, of which EUV power is only one. Even with a >1,000W source, High-NA adoption is likely to remain memory-led and layer-selective until DTCO, stitching and inspection ecosystems mature in parallel.

Stochastics: resolution gains expose photon / resist noise limits

In our view, stochastic variability is an under-appreciated but increasingly constraining challenge for High-NA EUV. At a basic level, stochastics refers to the random variability in where and how photons are absorbed in the resist, which at very small dimensions translates into line edge roughness, micro-bridging, line breaks or randomly failing vias. As critical dimensions shrink and k_1 is pushed closer to physical limits, these random effects cease to average out and instead become yield-relevant. High-NA improves optical resolution and image contrast, but it does not eliminate the discrete nature of EUV photons or resist chemistry. Our analysis points to a paradox: higher resolution can make stochastic failures more visible rather than less frequent.

This matters particularly for early High-NA layers, which tend to be the most aggressively scaled routing or via layers. These layers are already stochastic-limited under low-NA EUV, and High-NA pushes them further into regimes where dose, resist thickness and process latitude are tightly constrained. Simply increasing dose is not a free solution, as it hits throughput and cost of ownership. As a result, stochastics becomes a system-level optimisation problem involving resist chemistry, illumination, OPC, and inspection sensitivity, not just scanner optics.

That said, we do not see stochastics as a structural blocker to High-NA adoption. The industry has repeatedly solved similar issues - first in ArF immersion and later in low-NA EUV - through resist innovation, smarter pattern regularisation and selective layer deployment. In our view, stochastics is precisely why High-NA adoption will be slow and layer-selective rather than broad and immediate. It delays value realisation, but it does not invalidate the long-term logic of High-NA.

Inspection: High-NA raises the bar faster than inspection capacity scales

Inspection is, in our view, the most immediate operational constraint accompanying High-NA EUV. Moving to higher NA tightens tolerances across CDU, LCDU, overlay and defectivity, while simultaneously shrinking printable defect sizes. High-NA does not just require better lithography; it requires inspection tools that can reliably detect rarer, smaller and more stochastic defects without becoming a throughput bottleneck. This is particularly acute for logic, where yield sensitivity to local defects is high and dies are large.

The challenge is that inspection scaling lags lithography scaling by design. Optical wafer inspection struggles at sub-2nm dimensions, pushing fabs toward multi-beam e-beam inspection, which is slower, more expensive and harder to deploy at scale. At the mask level, High-NA re-elevates actinic inspection from “nice to have” to near-essential, as higher resolution and stronger mask 3D effects make non-actinic inspection insufficient. Our analysis suggests that inspection load per EUV layer rises meaningfully with High-NA, increasing both cycle time and capital intensity.

However, as with stochastics, we view inspection as a pacing item, not a dead-end. Tool vendors are investing heavily in multi-beam e-beam, AI-assisted defect classification and higher-throughput actinic mask inspection. Importantly, High-NA adoption itself creates the economic incentive to scale inspection capability. In our view, inspection constraints reinforce the case for a gradual, memory-led and layer-selective High-NA rollout - slow and uneven, but ultimately resolvable rather than structurally prohibitive.

Ecosystem maturity

High-NA EUV is best understood not as a scanner upgrade, but as a system-level transition spanning optics, metrology, inspection, masks, resists, and computational lithography. While progress has been made across the ecosystem, maturity is uneven. The most critical - and least replaceable - component is projection optics, supplied exclusively by Carl Zeiss SMT, followed closely by metrology and inspection infrastructure, which must operate at tighter tolerances than any previous lithography generation.

ZEISS optics: the structural bottleneck in High-NA EUV

The move from 0.33 NA to 0.55 NA fundamentally reshapes EUV optics. High-NA requires a complete redesign of the projection system, including:

- Much larger mirrors (metre-class dimensions),
- Higher asphericity and curvature, and
- Significantly tighter control of surface figure, roughness, and multilayer uniformity.

ZEISS is the sole supplier of EUV projection optics, which makes High-NA uniquely dependent on a single industrial capability. According to ZEISS technical disclosures, High-NA optics demand sub-nanometre control of figure and roughness across very large, heavy mirrors, with stability requirements that extend from static metrology into dynamic scanner operation (thermal load, vibration, and long-term drift).

At High-NA, optical errors that were previously second-order become yield-relevant:

- Aberrations directly impact overlay and CDU, not just resolution,
- Multilayer roughness contributes to image contrast loss and stochastic variation, and
- Transmission losses must be tightly managed to avoid throughput collapse at higher resolution.

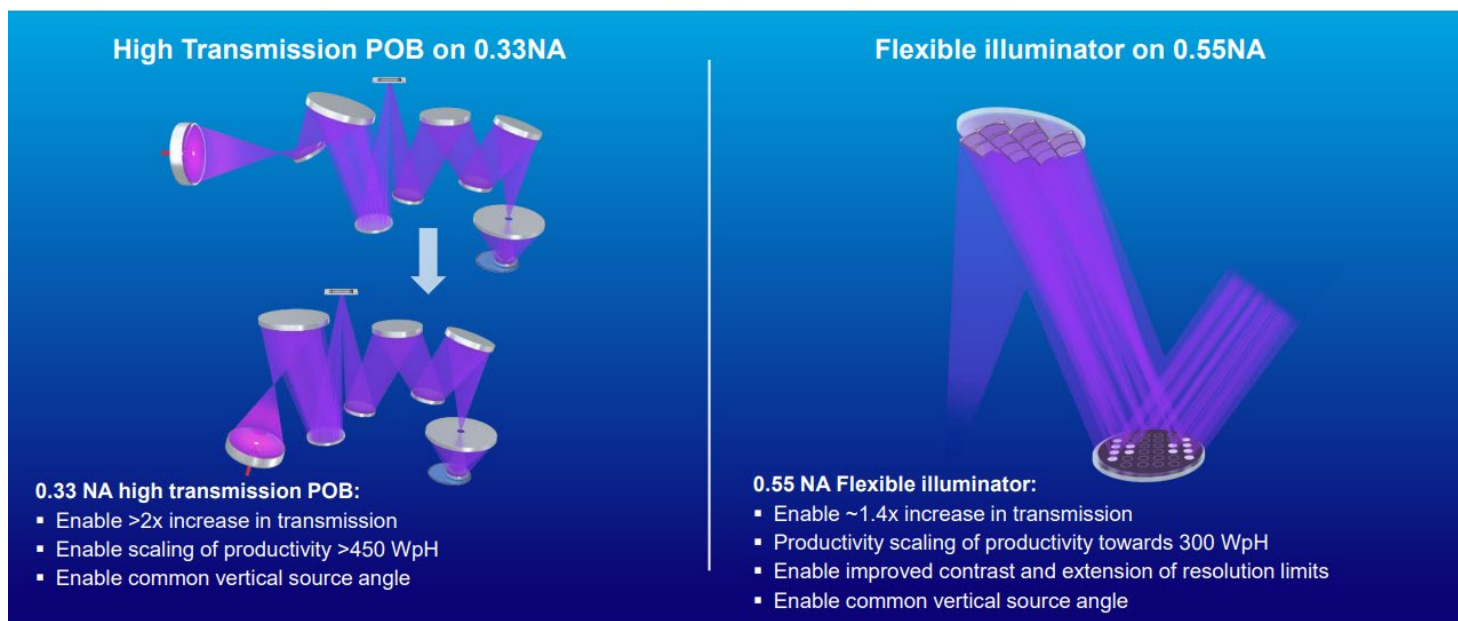
Importantly, optics manufacturing maturity is not only about demonstrating feasibility, but about repeatability and scaling. ZEISS has published that High-NA optics require entirely new manufacturing, polishing, coating, and metrology flows compared with prior EUV generations, and that optical metrology itself is a major development challenge due to the size, weight, and precision of the mirrors involved. This makes optics the long-pole in scaling High-NA system volumes, even after first systems are technically functional.

Exhibit 20: ASML's innovation in Optics alongside Zeiss is critical to high-NA adoption

Flexible illuminator will enable much higher transmission, driving productivity higher

High transmission optics for 0.33 NA and 0.55 NA enable a significant step in productivity

High transmission optics are enabled on the high productivity common platform



ASML

November 14, 2024

Page 32
Public

Source: Company report

BofA GLOBAL RESEARCH

Why TSMC is not committing

In our view, TSMC's reluctance to commit High-NA EUV to its near-term leading nodes is best understood as a DTCO return-on-complexity decision rather than a debate about optical capability. In public comments, TSMC has repeatedly stated that it does not require High-NA EUV for its next-generation process technologies (including A16 and A14), and that it will adopt High-NA only when it provides a "meaningful, measurable benefit." This aligns with a DTCO framing: if TSMC can continue delivering competitive PPA and density by co-optimising transistor architecture, standard cells, routing rules and process integration within the existing 0.33-NA EUV ecosystem, then High-NA's incremental benefit is not yet sufficient to justify the transition costs. Put simply, TSMC is signalling that Low-NA EUV + DTCO still has runway. The decision is therefore not about denying High-NA's physics; it is about maximising the value of the current design/manufacturing platform before introducing a technology that forces additional constraints into the design system.

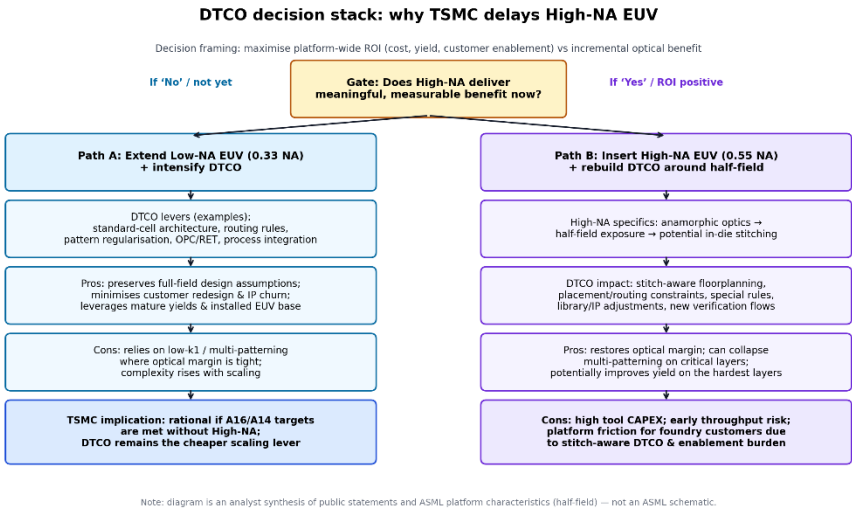
High-NA Increases Optical Freedom but Constrains the Design Platform

The critical DTCO link is that High-NA is not a "drop-in" upgrade. ASML itself explains that the High-NA EXE platform uses anamorphic optics and that the exposure field is half the size of current NXE systems. A half-field immediately raises the probability that large dies will need multiple exposures (i.e., in-die stitching), which is precisely where DTCO burden increases qualitatively: design rules, floor planning, placement and routing have to become stitch-aware rather than simply "lithography-friendly." From TSMC's standpoint as a platform foundry, this matters more than it would for an IDM, because TSMC's value proposition is broad customer enablement and IP reuse. A High-NA insertion that increases constraints (even if only on some layers) risks fragmenting the platform: more node-specific caveats, more re-validation, and potentially more customer redesign effort. That DTCO cost is paid upfront and across many customers, while the lithographic benefit is often layer-specific and may not translate into immediate, platform-wide economic advantage.

TSMC's High-NA Decision Is About Timing and Returns

Finally, TSMC's messaging also reflects capital discipline consistent with its DTCO strategy. Multiple recent reports cite TSMC executives confirming that the company will postpone using ASML's High-NA EUV tools for production until at least 2029, framing the tools as very expensive and emphasising the ability to keep scaling by leveraging existing EUV technology. The implication is that TSMC currently sees a higher ROI in continuing to extract scaling and cost improvements from the installed Low-NA base (and associated DTCO advances) than in accelerating a High-NA transition with its additional tool and integration costs. This is consistent with the "platform foundry" logic: TSMC will move when the benefits are sufficiently large and broadly applicable to justify the ecosystem shift, rather than moving early to prove technical leadership. In our own internal framing in our prior report [25 for 2026. Semis top pick. Litho intensity is ... rising](#), High-NA adoption is also modelled as gradual and customer-specific, reinforcing that the key variable is not feasibility but timing and ROI.

Exhibit 18: DTCO decision stack: why TSMC delays high-NA EUV
Decision framing maximise platform-wide ROI (cost, yield, customer enablement) vs incremental optical benefit



Source: BofA Global Research

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Why Intel, SK Hynix and Samsung are probably more willing

Why High-NA Economics Are Cleaner in Memory

The economic logic for earlier High-NA engagement is cleaner in memory than in leading-edge logic: higher resolution can directly translate into more bits per wafer and/or fewer patterning steps on the most critical layers, which improves both cost and yield. SK Hynix makes this case unusually explicitly in its own communications around installing a High-NA system for mass production. In its press release, the company defines High-NA EUV as a next-generation lithography system that “delivers much better resolution” via higher numerical aperture, and links “more sophisticated pattern” capability to producing “more chips from a wafer” and improved “power efficiency and performance.” It further states that with High-NA it plans to “simplify the existing EUV process and accelerate the development of the next-generation memory” to improve “product performance and cost competitiveness.” The critical point for your note is that these benefits are layer-specific but economically meaningful in DRAM: when a tighter pitch layer is approaching the manufacturability limits of Low-NA EUV, High-NA can (in principle) reduce reliance on multi-patterning, lower overlay accumulation and tighten edge placement control, which improves yield robustness on the layers that dominate bit density.

DRAM Largely Sidesteps the Stitching Penalty

DRAM is typically far less exposed to the stitching penalty that complicates High-NA adoption for large logic dies. High-NA’s anamorphic optics reduce field size and can require “in-die stitching” when a die is larger than the High-NA exposure field. SPIE material states the High-NA exposure field is $26 \times 16.5 \text{ mm}^2$, and notes that “in-die stitching may be required in order to create dies larger than [the] High-NA exposure field.” By contrast, a concrete example of a modern DRAM die size is much smaller: TechInsights reports a Micron 16Gb DDR5 DRAM die measuring roughly $6.35 \text{ mm} \times 5.67 \text{ mm}$ ($\sim 36 \text{ mm}^2$). A die of that footprint fits comfortably inside a $26 \times 16.5 \text{ mm}$ field, implying that typical DRAM dies can avoid in-die stitching altogether, removing one of the biggest “DTCO tax” items that affects high-performance logic SoCs. The nuance to keep in the note is: High-NA stitching is not “gone,” it is simply much less binding for DRAM because stitching is only required when die size exceeds the reduced field, and DRAM dies are often far smaller than the High-NA field.

Samsung’s Asymmetric High-NA Strategy

Samsung is more complex because the incentives differ across foundry vs memory. Public reporting suggests High-NA engagement begins with foundry R&D and evaluation for a 1.4nm-class process, which is consistent with a strategic “catch-up / differentiation” posture (tool leadership can help credibility when competing for advanced logic customers). At the same time, the same reporting indicates a more cautious stance for DRAM deployment timing, citing tool cost and roadmap considerations (including potential architecture transitions) - i.e., Samsung’s “keenness” is most visible in the foundry context, while memory insertion can remain selective and staged. This asymmetry is important to spell out: memory makers benefit more cleanly from resolution because it maps to cost-per-bit, while foundry adoption must also navigate platform-wide customer enablement friction; Samsung’s vertically integrated model can internalise more of that friction than a pure-play foundry, which helps explain why it can still be constructive on High-NA for certain use cases even if mass deployment is gradual.

Exhibit 19: High-NA EUV economics: Logic vs Memory

DRAM adoption of high-NA faces much lower hurdles than Logic

High-NA EUV Economics: Logic vs Memory

Dimension	Logic / Foundry	Memory (DRAM / HBM)
Typical die size vs High-NA field	Often larger than 26x16.5 mm → stitching likely	Typically much smaller → stitching usually avoided
DTCO burden	High: stitch-aware floorplanning, routing, libraries	Low: highly regular layouts
Benefit of higher resolution	Selective, layer-specific	Direct cost-per-bit and yield gain
Multi-patterning impact	Partial substitution; stitching offsets gains	Cleaner removal of EUV multi-patterning
Yield sensitivity	Design complexity + stitch regions	Pattern fidelity and repetition
Cost structure driver	CAPEX + enablement + customer redesign	Bits per wafer and yield
Customer enablement	Platform-wide and complex	Mostly internal and controlled
High-NA adoption logic	Wait until platform-wide ROI is clear	Earlier adoption justified by economics

Source: BofA Global Research estimates

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High-NA Makes Strategic Sense for Intel

Intel's earlier and more decisive engagement with High-NA EUV is economically and organisationally rational in our view, given its vertically integrated IDM model, smaller die insertion points, and strategic need to reset manufacturing leadership. Unlike a platform foundry, Intel can internalise the DTCO burden associated with High-NA, co-optimising design, process and manufacturing for a limited set of largely internal products rather than hundreds of external customers. This makes stitch-aware design rules, layer-selective constraints and early-learning inefficiencies more manageable. In addition, Intel's initial High-NA use cases are better aligned with the technology's constraints: server and client CPUs can be partitioned into smaller tiles, reducing exposure to in-die stitching, while High-NA can be targeted at the most resolution-limited layers rather than deployed node-wide. Strategically, High-NA also serves a different purpose for Intel than for TSMC: it is not about incremental platform optimisation, but about accelerating learning, signalling process leadership, and differentiating vs TSMC in advanced logic as Intel Foundry Services ramps. In this context, accepting lower early productivity and higher DTCO complexity is a rational trade-off, because the long-term payoff is faster maturity, credibility with prospective foundry customers, and a steeper internal learning curve ahead of broader industry adoption.

Glossary of technical terms

ArFi (Argon Fluoride Immersion lithography)

193-nm deep-ultraviolet lithography using immersion techniques to improve resolution. Still widely used in advanced nodes alongside EUV, particularly for less critical layers.

CDU – Critical Dimension Uniformity

CDU measures the variation of a nominal critical dimension (e.g. line width or space) across the wafer or within a die after lithography and etch. Poor CDU leads to device variability, leakage dispersion and performance degradation, particularly at advanced nodes where design margins are tight. CDU is typically driven by a combination of dose control, focus stability, mask effects, resist behaviour and etch sensitivity.

LDCU – Local Critical Dimension Uniformity

LDCU refers to short-range, local variations in critical dimensions, often measured over small spatial windows (e.g. within a cell, block, or neighbouring features). LDCU is closely linked to stochastic effects in EUV lithography, such as photon shot noise and resist chemistry randomness, and is increasingly a yield limiter at sub-3nm dimensions. While CDU captures global variation, LDCU is more relevant for local electrical performance and defectivity, especially for dense routing and via layers.

CPU – Central Processing Unit

The CPU is the primary general-purpose processor in a computing system, optimised for low-latency, sequential and control-intensive tasks. CPUs typically prioritise single-thread performance, complex instruction handling and cache coherence, making them central to operating systems, application logic and control workloads. In advanced logic manufacturing, CPUs are often large, complex designs with tight performance and yield sensitivity, which amplifies DTCO and stitching challenges at leading nodes.

GPU – Graphics Processing Unit

The GPU is a highly parallel processor designed to execute many simple operations simultaneously, making it well suited to graphics rendering, AI acceleration and high-throughput compute workloads. GPUs typically consist of large arrays of relatively uniform compute units and are increasingly implemented using chiplet-based architectures. From a manufacturing perspective, GPUs can be more amenable to tiling and partitioning than monolithic CPUs, which can partially mitigate High-NA stitching and DTCO constraints depending on design choices.

DRAM

Dynamic random access memory

DTCO (Design-Technology Co-Optimisation)

The tightly coupled optimisation of device architecture, process technology, and physical design rules. At advanced nodes, DTCO is the primary lever enabling scaling; High-NA increases DTCO complexity due to stitching-aware layouts.

DUV (Deep Ultraviolet lithography)

Lithography using wavelengths longer than EUV (193 nm ArF). Remains essential for non-critical layers and contributes meaningfully to ASML revenues despite leading-edge EUV adoption.

EUV (Extreme Ultraviolet lithography)

Lithography at a 13.5-nm wavelength used for printing the most critical layers at advanced nodes. ASML is the sole supplier of EUV scanners.

FinFET (Fin Field-Effect Transistor)

A 3D transistor architecture in which the channel is formed as a vertical “fin” surrounded by the gate on three sides. FinFETs replaced planar transistors at advanced nodes to improve electrostatic control, reduce leakage, and sustain scaling through the early EUV era. Most current production nodes still rely on highly optimised FinFET + DTCO rather than a transistor reset.

GAA (Gate-All-Around transistor)

A next-generation 3D transistor architecture in which the gate fully surrounds the channel (typically implemented as stacked nanosheets). GAA improves electrostatic control versus FinFETs and enables further scaling at sub-2nm-class nodes. Importantly for this note, GAA shifts more of the scaling burden to **DTCO and layout optimisation**, allowing foundries to delay High-NA EUV insertion by extracting additional value from Low-NA EUV combined with new device architectures.

High-NA EUV

Next-generation EUV lithography with a numerical aperture of ~0.55 (vs ~0.33 for today's Low-NA EUV). Offers materially higher resolution but introduces anamorphic optics, smaller exposure fields, stitching, and higher integration complexity, leading to selective and gradual adoption.

Low-NA EUV

Current production EUV scanners (NA ~0.33, e.g. NXE platforms). These tools remain the primary driver of ASML earnings through the forecast period due to rising volumes, higher layer counts, and improving productivity.

Installed Base Management (IBM)

ASML's services, upgrades, and productivity enhancements for the installed lithography fleet. IBM is structurally margin-accretive and benefits from the expanding EUV installed base.

k-factor (k_1)

A dimensionless process parameter in the Rayleigh equation that captures how aggressively lithography is pushed beyond basic optical limits. As k_1 approaches its physical minimum, resolution gains increasingly rely on new optics (e.g. High-NA) or DTCO.

Layer count (EUV layers)

The number of process layers in a chip manufactured using EUV lithography. Rising EUV layer counts - particularly in advanced logic and DRAM, are a key structural driver of lithography demand.

Logic (end-market)

Semiconductor devices such as CPUs, GPUs, AI accelerators and SoCs. Leading-edge logic often uses very large dies, making High-NA adoption more complex due to stitching and platform DTCO risk.

Memory (DRAM / HBM)

Semiconductor memory devices characterised by smaller die sizes and repetitive structures. High-NA adoption is cleaner economically for DRAM because smaller dies typically avoid in-die stitching and resolution maps directly to cost-per-bit improvements.

MMO – Mask Manufacturing Overlay

MMO refers to overlay errors introduced during mask fabrication, i.e. misalignment between successive pattern layers written on the reticle itself. These errors propagate directly to the wafer during exposure and therefore set a hard floor on achievable overlay and local pattern placement accuracy, independent of scanner performance. As feature sizes shrink and High-NA tightens overlay budgets, MMO becomes increasingly relevant because even small mask-side overlay errors can translate into yield- or performance-limiting defects on critical layers.

NA (Numerical Aperture)

A measure of an optical system's ability to capture diffracted light. Higher NA improves resolution but increases optical and manufacturing complexity.

OPC = Optical Proximity Correction

In semiconductor lithography, OPC refers to modifications made to the photomask layout so that the printed pattern on the wafer matches the intended design, despite optical distortions in the exposure process.

Overlay

The accuracy with which successive lithography layers are aligned. Overlay errors are particularly critical at stitch boundaries in High-NA EUV.

Productivity (wph / kWPM)

Throughput of a lithography tool, measured in wafers per hour (wph) or thousands of wafers per month (kWPM). Productivity improvements are a major lever supporting Low-NA EUV earnings growth.

Reticle

The photomask containing the chip pattern. Reticle size limits single-exposure die dimensions; High-NA's reduced exposure field increases the likelihood of in-die stitching for large dies.

Stitching

The process of combining multiple exposure fields to pattern a single die. In-die stitching introduces yield and design risks and is the single largest structural headwind for High-NA adoption in large-die logic. [high-NA note | Word]

Stochastics

Random variability in photon absorption and resist chemistry at very small dimensions, leading to defects such as line-edge roughness or via failures. Becomes more visible at higher resolution and constrains aggressive High-NA deployment.

WFE (Wafer Fab Equipment)

Capital equipment used in semiconductor manufacturing. Lithography is the most critical and capital-intensive segment within WFE.

WSPM (Wafers per month)

A measure of wafer fabrication capacity. Translating WSPM additions into EUV exposure demand underpins the bottom-up shipment model used in the note.

Price objective basis & risk

ASML Holding N.V. (ASMLF / ASML)

We have set a PO of €1,710 (\$2,019 for the ADR). We use a 30.0x CY27E EV/EBITDA to derive our PO, within its historical range of 18x to 34x and above its 5-year historical median of 26x given accelerated EPS growth

Upside risks to our PO are:

- 1) Higher-than-expected demand for EUV and DUV machines
- 2) Higher-than-expected gross margins on EUV machines and services
- 3) Stronger-than-expected Semi Capex environment driven by macroeconomic trends and higher capital intensity per wafer.

Downside risks to our PO are:

- 1) Delays in ramping volume shipments of EUV machines
- 2) Lower-than-expected gross margins on EUV machines
- 3) Weaker-than-expected Semi Capex driven by macro slowdown/oversupply
- 4) China export restrictions

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EMEA - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
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	ASM International N.V.	ASMXF	ASM NA	Didier Scemama
	ASML Holding N.V.	ASMLF	ASML NA	Didier Scemama
	ASML Holding N.V.	ASML	ASML US	Didier Scemama
	BE Semiconductor Industries N.V.	BESVF	BESI NA	Didier Scemama
	Infineon Technologies AG	IFNMF	IFX GY	Didier Scemama
	Infineon Technologies AG	IFNNY	IFNNY US	Didier Scemama
	Nokia	NOKBF	NOKIA FH	Oliver Wong
	Nokia	NOK	NOK US	Oliver Wong
	Nordic Semiconductor	NDCVF	NOD NO	Oliver Wong
	VAT Group AG	VTTGF	VACN SW	Oliver Wong
NEUTRAL				
	Logitech International S.A.	XLGKF	LOGN SW	Didier Scemama
	Logitech International S.A.	LOGI	LOGI US	Didier Scemama
	Soitec	SLOIF	SOI FP	Oliver Wong
	STMicroelectronics NV	STMEF	STMPA FP	Didier Scemama
	STMicroelectronics NV	STM	STM US	Didier Scemama
	Technoprobe S.p.A.	XMDDF	TPRO IM	Oliver Wong
UNDERPERFORM				
	ams Osram AG	AUKUF	AMS SW	Didier Scemama
	Comet Holding AG	XCHHF	COTN SW	Oliver Wong
	Ericsson	ERIXF	ERICB SS	Oliver Wong
	Ericsson	ERIC	ERIC US	Oliver Wong
	Inficon Holding AG	IFCND	IFCN SW	Oliver Wong
	Melexis	MLXSF	MELE BB	Amelia Banks
	Mycronic AB	MICLF	MYCR SS	Oliver Wong
	Siltronic AG	SLLF	WAF GR	Didier Scemama

iQmethodSM Measures Definitions

Business Performance

Return On Capital Employed

Return On Equity

Operating Margin

Earnings Growth

Free Cash Flow

Quality of Earnings

Cash Realization Ratio

Asset Replacement Ratio

Tax Rate

Net Debt-To-Equity Ratio

Interest Cover

Valuation Toolkit

Price / Earnings Ratio

Price / Book Value

Dividend Yield

Free Cash Flow Yield

Enterprise Value / Sales

EV / EBITDA

Numerator

NOPAT = (EBIT + Interest Income) × (1 – Tax Rate) + Goodwill Amortization

Net Income

Operating Profit

Expected 5 Year CAGR From Latest Actual

Cash Flow From Operations – Total Capex

Numerator

Cash Flow From Operations

Capex

Tax Charge

Net Debt = Total Debt – Cash & Equivalents

EBIT

Numerator

Current Share Price

Current Share Price

Annualised Declared Cash Dividend

Cash Flow From Operations – Total Capex

EV = Current Share Price × Current Shares + Minority Equity + Net Debt +

Other LT Liabilities

Enterprise Value

Denominator

Total Assets – Current Liabilities + ST Debt + Accumulated Goodwill

Amortization

Shareholders' Equity

Sales

N/A

N/A

Denominator

Net Income

Depreciation

Pre-Tax Income

Total Equity

Interest Expense

Denominator

Diluted Earnings Per Share (Basis As Specified)

Shareholders' Equity / Current Basic Shares

Current Share Price

Market Cap = Current Share Price × Current Basic Shares

Sales

Basic EBIT + Depreciation + Amortization

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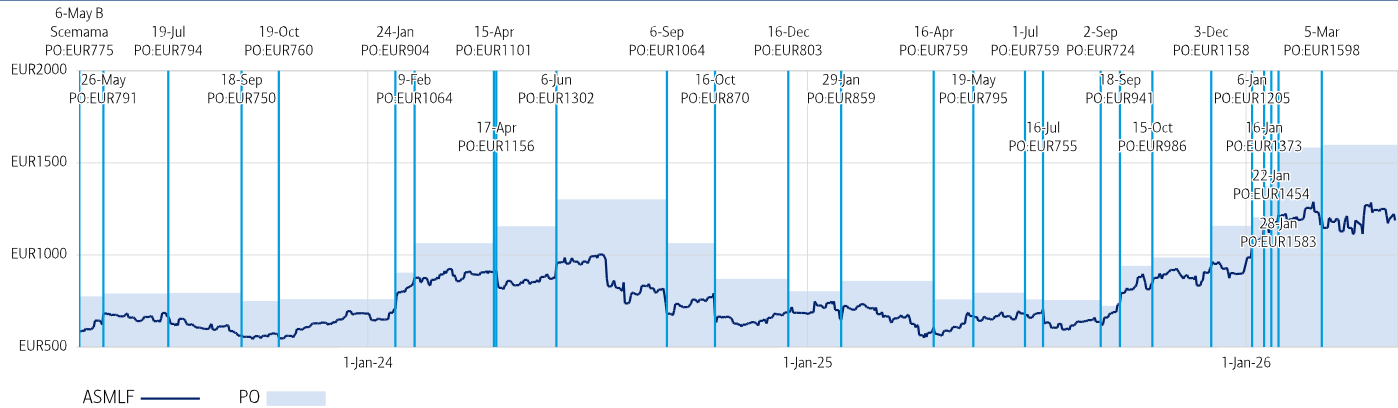
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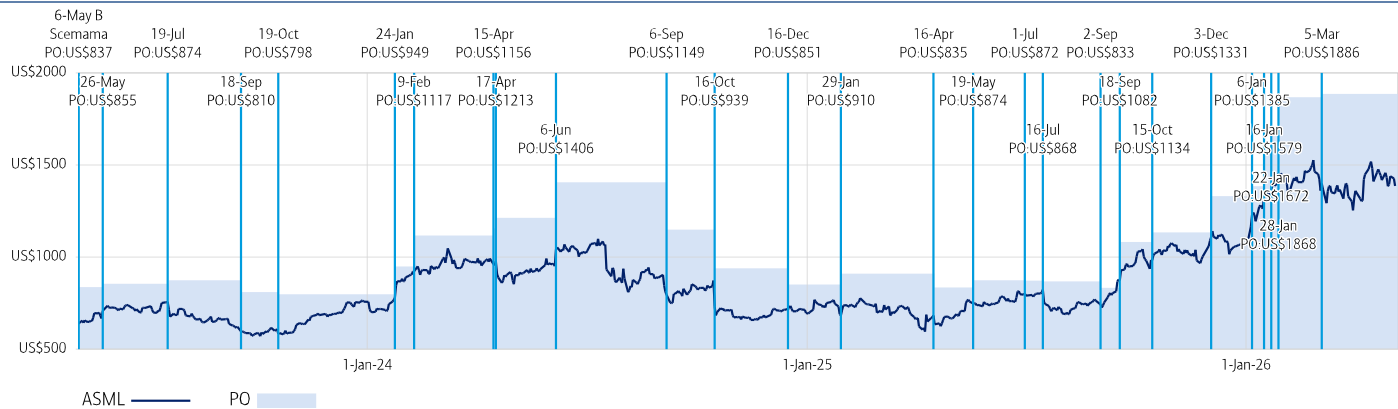
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ASML (ASMLF) Price Chart



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ASML (ASML) Price Chart



The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Technology Group (as of 31 Mar 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	234	58.50%	Buy	123	52.56%
Hold	90	22.50%	Hold	43	47.78%
Sell	76	19.00%	Sell	23	30.26%

Equity Investment Rating Distribution: Global Group (as of 31 Mar 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	1993	55.76%	Buy	1186	59.51%
Hold	821	22.97%	Hold	509	62.00%
Sell	760	21.26%	Sell	400	52.63%

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster ^{R2}
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

^{R2}Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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